FF-H-111a
July 8, 1948
SUPERSEDING
Fed. Spec. FF-H-111
September 5, 1933

FEDERAL SPECIFICATION

HARDWARE, BUILDERS'; SHELF AND MISCELLANEOUS

This specification was approved by the Director, Bureau of Federal Supply, for the use of all departments and establishments of the Government.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—The hardware covered by this specification is that generally classified as shelf and miscellaneous builders' hardware. It is not intended for marine use. (See definition, 6.5, and index, page 49.)

1.2 Types.—The hardware shall be of the types described in section 3, as specified.

2. APPLICABLE SPECIFICATIONS

2.1 Specifications.—The following Federal Specifications, of the issues in effect on date of invitation for bids, form a part of this specification:

L–P–406—Plastics, Organic; General Specifications, Test Methods.

QQ–M–151—Metals; General Specification for Inspection of.

QQ–Z–363—Zinc-Base Alloy; Die Castings.

2.2 Specifications and publications applicable only to individual departments are listed in section 7.

3. REQUIREMENTS

CONTENTS OF SECTION 3

	Paragraph
Applicable requirements, options	3.1—3.1.1
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Finishes	3.6—3.14
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Types of hardware	3.15.4 (pages 5 to 38)

3.1 Applicable requirements.—The hardware shall conform (a) to all requirements given herein under the type number for which it is furnished, (b) to any other applicable requirements in this section, and (c) to any additional requirements specified in the invitation for bids in accordance with selective variations provided herein as to size, length, finish, etc. The hardware shall also meet applicable tests in section 4.

3.1.1 *Supplier's options.*—Where variations are permitted herein as to materials, construction, appearance, or other characteristics, hardware conforming to any of the options thus provided, as well as to all other applicable requirements given herein, shall be considered as meeting this specification.

3.2 Materials, general.

3.2.1 *Optional materials.*—Where materials specified herein are designated as given below, the respective optional materials listed

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may be substituted. (This option does not apply to finish. See 3.6.1.)

Designated material:	Optional material
Brass	Bronze. (See 3.3.1.)
Brass, nickel plated	White bronze (unplated).
Bronze	(See 3.3.2.)
Steel	Wrought iron.
Cast iron	Forged iron; cast, rolled, or extruded aluminum alloy.
Cast (nonferrous metal)	(See 3.2.2.)
Wrought (metal)	Rolled, forged, extruded, or pressure-cast form of the metal specified.

3.2.2 *Substitution of wrought nonferrous metal for cast nonferrous metal.*—Where cast nonferrous metal is specified, forged or extruded forms of the same nonferrous metal may be furnished, provided the forged or extruded items or parts are of substantially the same thickness or weight as commercially equivalent cast items or parts.

3.2.3 *Substitution of ferrous metal for nonferrous metal.*—When black finishes US18 and US18A are applied to hardware of types given herein which specify bronze or other nonferrous metal, the hardware shall be of cast iron in lieu of cast nonferrous metal, and of wrought iron or steel in lieu of wrought nonferrous metal; except that in certain parts of locks nonferrous metal shall be furnished as follows: fronts, strikes, and cylinder heads shall be of brass or bronze with dull black finish US19; latches, dead bolts, stops, and cylinder plugs shall be of bronze with bright finish. (See 3.11 and 3.12.)

3.3 Brass and bronze.

3.3.1 *Chemical composition.*—Brass and bronze shall be suitable alloys composed chiefly of copper, with zinc and other elements as needed to provide the desired appearance and other properties usually associated with red bronze and yellow brass.

3.3.2 *Bronze.*—Where bronze is specified herein as a material, without other requirements as to appearance or composition, brass may be furnished as an optional material. (This option does not apply to finish. See 3.6.1.)

3.3.3 *White bronze.*—The chemical composition of white bronze, except in keys, shall conform to table I. This material is also known commercially as "nickel bronze", "nickel silver", and "white metal".

TABLE I.

Chemical composition of white bronze

Element	Cast		Wrought	
	Minimum	Maximum	Minimum	Maximum
Copper	62	—	62	—
Nickel	19	—	17	—
Tin	2.5	—	—	—
Lead	—	6	—	1.5
Zinc	—	10	—	18
Manganese	—	1	—	0.5
Iron and other impurities	—	1	—	1

3.4 **Zinc-base-alloy die castings.**—The castings shall conform to Federal Specification QQ-Z-363, and shall be of either composition specified therein.

3.5 Plastics.

3.5.1 *Workmanship.*—Hardware items of plastic materials shall have clean, smooth surfaces free from warpage and rough edges. Exposed surfaces shall be uniform in gloss. There shall be no chipping or visible fracture of molded material where the "gate" is cut.

3.5.2 *Deformation in hot water.*—Hardware items of plastic materials, or test pieces cut therefrom, shall maintain their shape without appreciable warping, bending, or contraction due to unmolding, when immersed for 30 minutes in water kept at 160° ± 2°F. (See test, 4.8.1.)

3.5.3 *Accelerated aging test.*—Hardware items of plastic materials, or test pieces cut therefrom, shall not undergo appreciable changes in physical characteristics after being subjected to the accelerated aging treatment specified in 4.8.2.

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3.5.4 *Flammability.*—Hardware items of plastic materials, or test pieces cut therefrom, shall not burn at a rate greater than 3.0 inches per minute when tested as described in 4.8.3.

3.5.5 *Colorfastness.*—Hardware items of plastic materials shall show no streaks of discoloration after 50 hours exposure in the test for colorfastness to light specified in 4.8.4.

3.6 Finishes, general.

3.6.1 *Standard finishes.*—Exposed surfaces of hardware shall have such of the applicable standard finishes as are specified in the invitation for bids. (See 6.3.1 and table II, standard finishes); otherwise, the regular finishes specified in 3.6.2, or elsewhere herein, shall be furnished.

3.6.2 *Regular finishes.*—When no other finishes are specified herein or in the invitation for bids, brass shall have US3 finish, bright brass; bronze shall have US9 finish, bright bronze; and iron and steel shall be plated and have US9 finish. When unfinished iron or steel is specified herein, it may be cadmium-plated, zinc-coated, japanned, coppered, or unfinished. Iron and steel shall be free from rust.

3.6.3 *Lacquer.*—All natural color and plated hardware shall have a coating of transparent lacquer, unless otherwise specified herein; except that chromium-plated, nickel-plated, cadmium-plated, zinc-plated, and zinc-coated finishes, and white bronze, may be furnished without lacquer.

3.6.4 *Finishes on bronze.*—Bronze with bright plated and natural color finishes shall have highly reflective surfaces showing maximum smoothness, evenness, and uniformity, such as produced by grinding, polishing, and buffing, except for unpolished areas used for decorative effect; bronze with dull plated and natural color finishes shall be produced similarly and be uniform in texture. Brass furnished in lieu of bronze (see 3.3.2) shall also conform to the above requirements.

3.7 Nickel plating, finishes US14, US15.

3.7.1 *Nickel plating on brass and bronze.*—Nickel plating shall be applied directly to brass and bronze. On significant surfaces,[1] the minimum thickness of the nickel coating shall be at least 10 hundred-thousandths inch (0.00010 inch).

3.7.2 *Nickel plating on iron and steel.*—The nickel plating shall be applied either directly or over a coating of copper. On significant surfaces,[1] the minimum thickness of the nickel coating or of the combined layers of copper and nickel shall be at least 4 ten-thousandths inch (0.0004 inch), and when copper is used, the minimum thickness of the final nickel layer shall be at least 2 ten-thousandths inch (0.0002 inch). The plated coatings, after 16 hours continuous exposure to the salt spray test specified in Federal Specification QQ-M-151, shall show no appreciable corrosion on significant surfaces.

3.7.3 *Nickel plating on zinc-alloy die castings.*—The nickel plating shall be applied over a coating of copper. On significant surfaces,[1] the minimum thickness of the combined layers of copper and nickel shall be at least 5 ten-thousandths inch (0.0005 inch), the minimum thickness of the copper layer shall be at least 2 ten-thousandths inch (0.0002 inch), and the minimum thickness of the final nickel layer shall be at least 3 ten-thousandths inch (0.0003 inch). The plated coatings, after 16 hours continuous exposure to the salt spray test specified in Federal Specification QQ-M-151, shall show no appreciable corrosion on significant surfaces.

3.8 Chromium plating, finishes US26, US26D.

3.8.1 Chromium plating shall be applied over a coating of nickel as specified under 3.7 for the respective materials. On significant surfaces,[1] the minimum thickness of the chromium shall be at least one hundred-thousandths inch (0.00001 inch). Chromium plating applied over nickel plating on iron,

[1]For definition of significant surfaces, see 4.2.3.

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steel, and zinc-alloy die castings, after 16 hours continuous exposure to the salt spray test specified in Federal Specification QQ–M–151, shall show no appreciable corrosion on significant surfaces.

3.9 Cadmium plating, finish US2C.

3.9.1 On significant surfaces,[1] the minimum thickness of the cadmium plating shall be at least 15 hundred-thousandths inch (0.00015 inch). Zinc coating conforming to this requirement may be furnished in lieu of cadmium. (See 3.10.2.)

3.10 Zinc coating, finish US2H.

3.10.1 On significant surfaces,[1] the minimum thickness of the zinc coating shall be at least 5 ten-thousandths inch (0.0005 inch).

3.10.2 *Application of zinc coating.*—Zinc coating (galvanizing) may be applied either by hot dipping in molten zinc, or by electroplating. However, hot dipping shall not be employed for coating alloy or heat-treated steel that may be injuriously affected at the temperature of the molten zinc; for such steel, the coating shall be applied by electroplating.

3.11 Black finish US18, Bower-Barff, shall be applied only to iron and steel, and shall be produced by a suitable heat treatment of the metal, following which a coating of suitable oil, wax, or lacquer shall be applied. (See 3.2.3 and 6.2.)

3.12 Black finish US18A, sanded[2] rust-resisting black, shall be applied only to iron and steel. It shall consist of one or more coatings of black lacquer applied after the surface of the iron or steel is prepared by a commercial phosphate treatment, so as to produce an adherent corrosion-resisting base for the lacquer. The phosphate treatment shall be applied either directly to the iron or steel or over a zinc coating. (See 3.2.3 and 6.2.)

3.13 Japan coating, finishes US1B, US1D.—Japan coating shall adhere tightly to the metal surfaces and be dry and elastic. It shall not be subject to cracking, scratching, or flaking. The surfaces shall be black and, for finish US1B, shall show a high gloss.

3.13.1 Japan coating shall be considered dry when the pressure that can be exerted between the thumb and forefinger does not leave a mark which remains noticeable after the spot is lightly polished, as determined at room temperature (about 20°C., 68°F.).

3.14 Anodic film for aluminum.—Items for which aluminum with anodic film is specified herein shall be so treated by either the chromic acid or sulfuric acid process as to produce a uniform, continuous film of aluminum oxide, free from breaks, scratches, or other damage. The film shall withstand exposure for a period of 250 hours in the salt spray test described in Federal Specification QQ–M–151. At the end of the period of exposure, there shall be no evidence of corrosive attack, except that the presence of minute, isolated pits shall not be sufficient cause for rejection.

3.15 Workmanship.

3.15.1 All hardware shall be free from imperfections and defects which affect its appearance, durability, and serviceability.

3.15.2 All finished parts of hardware that are to be set flush shall have accurately machined, true edges. Edges of unfinished steel and iron, and unpolished work, need not be machine-finished but shall be smooth and reasonably true.

3.16 Hardware types.

3.16.1 *Fastenings.*—Each piece of hardware shall be furnished complete with all necessary screws, bolts, etc., for proper application. Such fastenings shall be of suitable size and type, and shall harmonize with the hardware as to material and finish.

[1] For definition of significant surfaces, see 4.2.3.

[2] This finish is described as "sanded" because of its texture, the appearance of which may be produced by sandblasting or by other suitable methods.

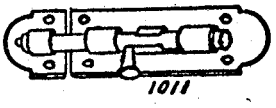
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3.16.2 *Illustrations.*—The illustrations shown herein are descriptive of the desired types, and shall not be construed as showing required designs; however, items of designs other than that illustrated and described shall be full commercial equivalents of the one shown. In general, the illustrations precede the descriptions. Numbers on each illustration correspond with numbers of the type, or first of a group of types, to which it applies.

3.16.3 *Dimensions and weights.*—The dimensions and weights given herein are descriptive of the desired types and sizes, and shall be considered approximate, unless otherwise specified herein; however, if dimensions and weights are not as given herein, the article shall be a full commercial equivalent of articles that meet all requirements.

3.16.4 *Requirements for individual types* shall be as follows:

BARREL BOLTS:



Type 1011.—Wrought brass. Minimum thickness of backplate 0.051". Lengths of bolt, 3", 4", and 5".

Type F1011.—Wrought steel, japanned finish. Otherwise the same as type 1011.

Type F1011C.—Wrought steel, cadmium- or zinc-plated (par. 3.9.1). Otherwise the same as type 1011.

BARREL BOLTS—HEAVY:

Type 1012.—Wrought brass. Minimum thickness of backplate 0.070". Lengths of bolt, 4", 6", and 8".

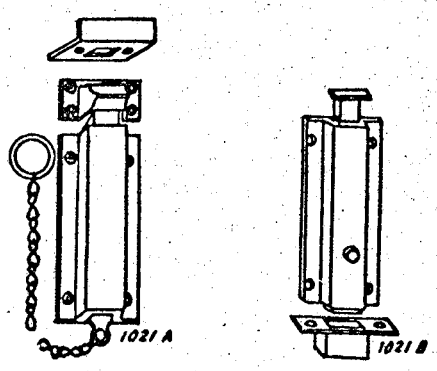
Type F1012.—Wrought steel, japanned finish. Otherwise the same as type 1012.

Type F1012C.—Wrought steel, cadmium- or zinc-plated (par. 3.9.1). Otherwise the same as type 1012.

BARREL BOLTS—HEAVY, CONT'D.:

Type 1013.—Cast bronze. Lengths of bolt, 4" and 5".

CHAIN AND FOOT BOLTS:

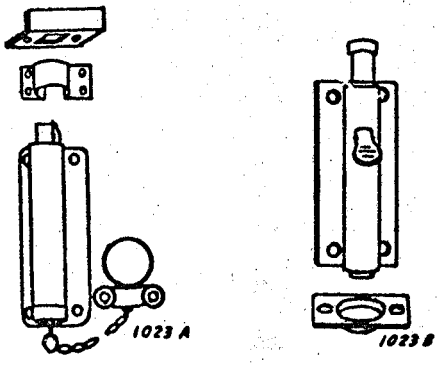


Type 1021A.—Chain bolt. Cast bronze. Lengths of case, 4" and 6". Regular and angle (reverse bevel) strikes.

Type F1021A.—Chain bolt. Cast iron. Otherwise the same as type 1021A.

Type 1021B.—Foot bolt. Cast bronze. Lengths of case, 4" and 6".

Type F1021B.—Foot bolt. Cast iron. Otherwise the same as type 1021B.



Type 1023A.—Chain bolt. Wrought brass. Round or square case. Regular and angle (reverse bevel) strikes. Dimensions as follows:

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CHAIN AND FOOT BOLTS, CONT'D.:

Dimensions for type 1023A

Lengths of bolt, inches	3	6
Minimum thickness of plate, inches	0.040	0.064

Type F1023A.—Chain bolt. Wrought steel, japanned finish. Round or square case. Regular and angle (reverse bevel) strikes.

Lengths of bolt, inches	3	6	8	10
Minimum thickness of plate, inches	0.040	0.064	0.064	0.072

Type F1023AC.—Chain bolt. Wrought steel, cadmium- or zinc-plated (par. 3.9.1). Otherwise the same as type F1023A.

Type 1023B.—Foot bolt. Wrought brass. Round or square case.

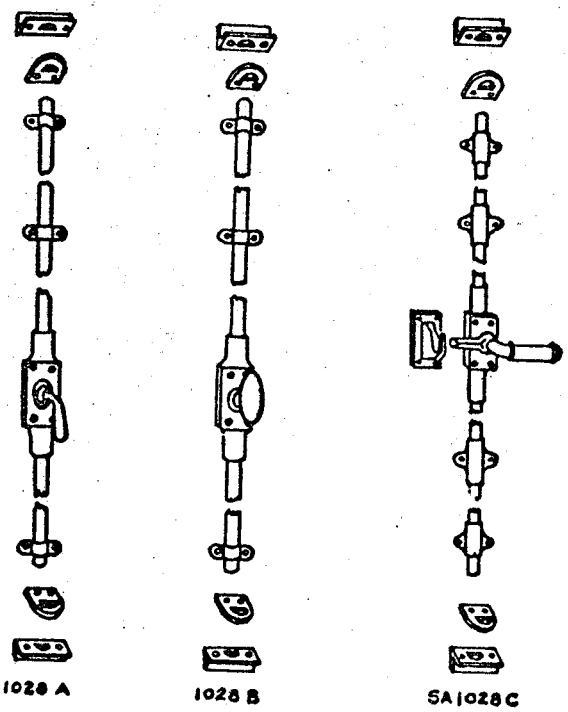
Lengths of bolt, inches	3	6
Minimum thickness of plate, inches	0.040	0.064

Type F1023B.—Foot bolt. Wrought steel, japanned finish. Round or square case.

Lengths of bolt, inches	3	6	8	10
Minimum thickness of plate, inches	0.040	0.064	0.064	0.072

Type F1023BC.—Foot bolt. Wrought steel, Cadmium- or zinc-plated (par. 3.9.1). Otherwise the same as type F1023B.

CREMONE BOLTS:



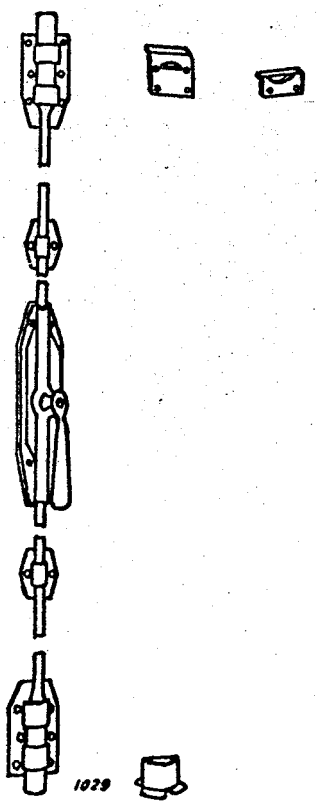
CREMONE BOLTS, CONT'D.:

Type 1028A.—With lever handle. For doors or sash opening out. Cast bronze, except wrought bronze rods and strikes. Half-oval or half-round bronze rod 5/8" wide. Guides for every 2 feet of rod. Operation from inside only, by quarter or half turn. Constructed so as not to deface paint in operating.

Type 1028B.—With oval knob. For doors or sash opening in. Otherwise the same as type 1028A.

Type SA1028C.—(For special application, see 6.2.) Same as type 1028A, except with center catch and rim center strike for double doors or sash. Reversible, or hand as specified.

GARAGE TY E:



Type F1029.—Wrought steel, extra heavy; japanned finish, except rods cadmium- or zinc-plated (par. 3.9.1) and without japan. 1/2" round or 3/4" half-oval rods, and round

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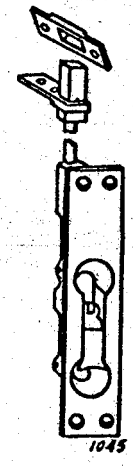
CREMONE BOLTS, CONT'D.:

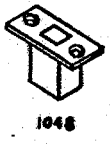
GARAGE TYPE, CONT'D.:

or half-oval bolt heads. Center guides for every 2 feet of rod. Angle and universal top strikes; bottom strike for concrete. Operation from inside only. Shall have automatic catch or other suitable means to retain handle and bolts in open position.

NOTE: When two bolts are for use in pairs, opposite hands should be specified.

LEVER EXTENSION FLUSH BOLTS:





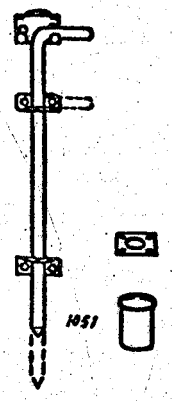
Type 1045.—Cast bronze plate 6⅜" x 1¼" minimum. Bolt head, brass, ½" square. Length of rod 12".

Type 1045R.—For rabbeted doors. Otherwise the same as type 1045.

Type 1046.—Cast bronze plate 5⅞" x ¾". Bolt head, brass, ⅜" diameter. Length of rod 12". Suitable for 1⅜" doors.

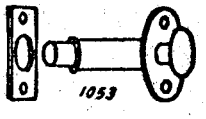
Type 1048.—Cast bronze dustproof strike to suit type of bolt specified. Approximately 1" x 2½" on face, by 2" depth; back fully enclosed.

CANE BOLT:



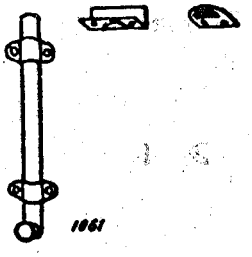
Type F1051.—Heavy wrought steel. Cadmium- or zinc-plated rod, minimum diameter ⅝". Guides with japanned finish. Strikes for wood and for concrete. Suitable stops shall prevent bolt from slipping out of guides. Minimum throw 6½". Lengths of bolt, 18" and 24".

MORTISE DOOR BOLT:



Type 1053.—Wrought bronze with steel bolt. Diameter of case approximately $\frac{5}{16}$"; backset minimum 1⅝".

SURFACE BOLTS:



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SURFACE BOLTS, CONT'D.:

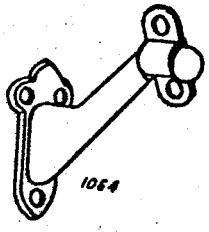
Type 1061.—Wrought bronze. ½" half-round or oval rod. ⅞" throw. Two guides, which shall prevent bolt from touching woodwork. Bolts shall not rotate, and shall be secure in both extended and withdrawn positions. Furnish with angle and universal strikes. Lengths 6" and 9".

Type F1061.—Wrought steel. Otherwise the same as type 1061.

Type 1062.—Wrought bronze. ⅜" half-round or oval rod. Lengths 4" and 6". Otherwise the same as type 1061.

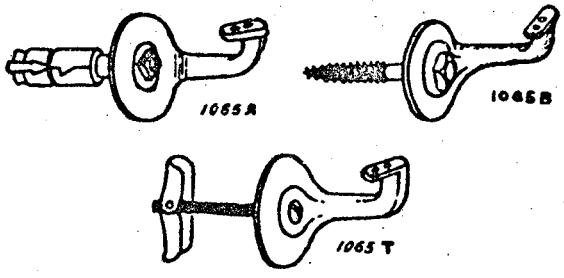
Type F1062.—Wrought steel. ⅜" half-round or oval rod. Lengths 4" and 6". Otherwise the same as type 1061.

HAND RAIL BRACKETS:



Type 1064.—Cast bronze. Adjustable. Minimum projection 2⅝" from base to center of rail.

Type F1064.—Cast iron. Otherwise the same as type 1064.



Type 1065A.—Cast bronze. Adjustable. 3" diameter round base. Minimum projection 3" from base to center of rail. Applied to wall with ½" x 3" wrought bronze machine bolt and suitable expansion sleeve.

HAND RAIL BRACKETS, CONT'D.:

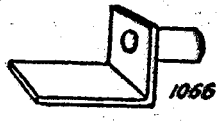
Type F1065A.—Cast iron, with steel machine bolt. Otherwise the same as type 1065A.

Type F1065B.—Cast iron. Applied to wall with ½" x 3" steel lag screw. Otherwise the same as type 1065A.

Type 1065T.—Cast bronze. Applied to wall with wrought bronze toggle bolt. Otherwise the same as type 1065A.

Type F1065T.—Cast iron. Applied to wall with steel toggle bolt. Otherwise the same as type 1065A.

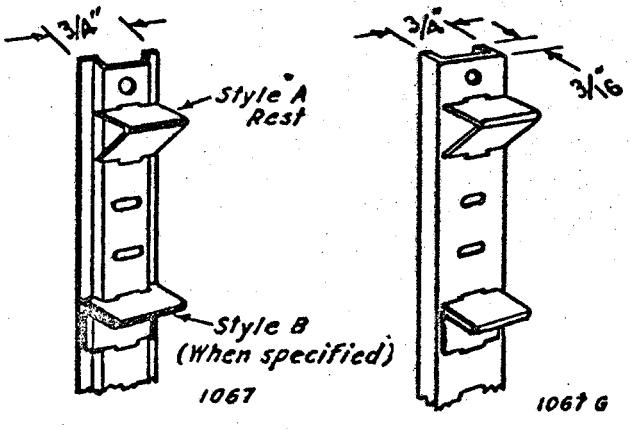
SHELF RESTS:



Type F1066.—Wrought steel, nickel-plated. Length of pin, ⅜"; diameter, ¼". Overall length approximately 1¼".

Type F1066A.—Same as type F1066, except cadmium-plated.

ADJUSTABLE SHELF SUPPORTS:



Type 1067.—For surface application. Wrought brass, nickel-plated. Screw fastenings for standard spaced not over 6 inches. Rests shall hook securely into slots in standard. Shall provide adjustment of shelf by ½" increments. Alternate slots shall be numbered in groups of not less

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ADJUSTABLE SHELF SUPPORTS, CONT'D.:

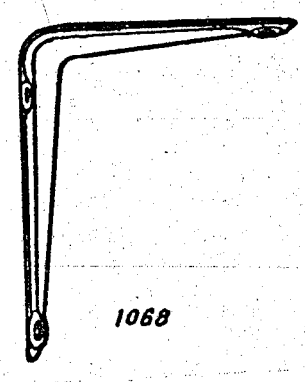
than 6. Minimum projection of rest, ¾"; minimum width, ¾". Minimum thickness of standard and rest, 0.048". Unless otherwise specified, furnish standard in 12-foot lengths, with one style "A" rest per foot.

Type F1067.—For surface application. Wrought steel, nickel-plated. Otherwise the same as type 1067.

Type 1067G.—For flush application in groove. Wrought brass. Same as type 1067, except formed for flush mounting.

Type F1067G.—For flush application in groove. Wrought steel. Formed for flush mounting. Otherwise the same as type 1067.

SHELF BRACKETS:



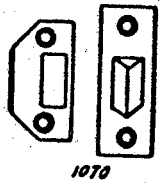
1068

Type F1068.—Wrought steel, japanned finish. One-piece tapered rib construction. Sizes, thickness, and screws as follows:

Dimensions and screws for type F1068

Size	Thickness of plate, minimum	Shelf plate screws, minimum			Wall plate screws, minimum		
		No.	Length	Size No.	No.	Length	Size No.
Inches	*Inch*		*Inch*			*Inches*	
4x5	0.035	3	¾	6	3	1¼	6
5x6	.035	3	¾	6	3	1¼	6
6x8	.047	3	¾	6	3	1¼	6
8x10	.062	3	¾	7	3	1¼	7
10x12	.062	3	¾	7	3	1¼	7
12x14	.068	3	¾	8	3	1¼	8

FRICTION CATCHES:



1070

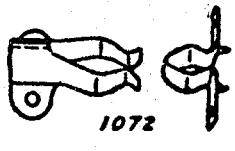
Type 1070.—Wrought brass. Spring action plunger. Size 2" x ¾". For application in mortise.



1071

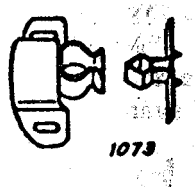
Type 1071.—Wrought brass tubular housing, ⅜" diameter x ½" deep. Spring action polished steel rounded plunger or ball, plated. Rounded concave strike of wrought steel, plated; ⅝" diameter. For application in bored holes.

FRICTION CATCHES, CONT'D.:



1072

Type F1072.—Tempered steel one-piece spring clip, ½" to 1" x 1⅜", having two prongs engaging wrought steel strike approximately ½" x 1⅜". For surface mounting. Cadmium- or zinc-plated, except that spring may have blued finish.



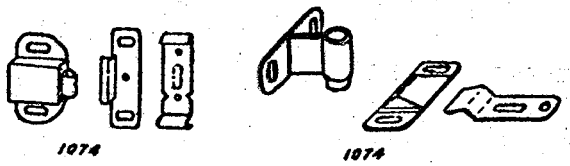
1073

(Continued on next page)

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FRICTION CATCHES,[1] CONT'D.:

Type F1073.—Steel. Housing approximately ½" x ½" with base 1¼" x ¾". Elongated holes for surface mounting. Two floating metal prongs held by spring tension to engage strike mounted on door. Strike approximately ⅜" x 1⅛". Cadmium- or zinc-plated, unless specified to be nickel-plated.

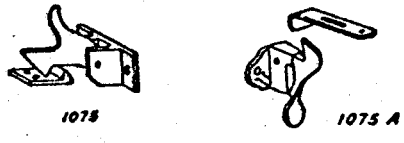

1074 1074

Type 1074.—Wrought brass. Housing with base approximately 1¼" x ⅝". Elongated holes for surface mounting. Strikes for shelf and door jamb; catch for mounting on inside of flush and lipped doors. Spring action plunger of oil-impregnated hardwood or of metal. Optional designs are illustrated.

Type F1074C.—Wrought steel, cadmium- or zinc-plated. Otherwise the same as type 1074.

Type F1074D.—Wrought steel, nickel-plated. Otherwise the same as type 1074.

ELBOW CATCHES:


1075 1075 A

Type 1075.—Cast brass, unpolished. Spring action catch. Size ⅝" x 1¼".

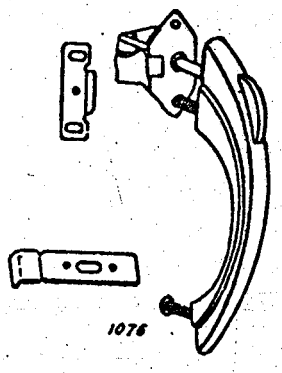
Type F1075.—Cast iron, unpolished. Spring action catch. Size ⅝" x 1¼".

Type F1075A.—Wrought steel, nickel-plated. Spring action catch. Size approximately 1" x $1\frac{5}{16}$".

Type F1075AC.—Wrought steel, cadmium- or zinc-plated (par. 3.9.1). Otherwise the same as type F1075A.

[1]Roller latches for regular doors are covered in FF-H-106, type 193.

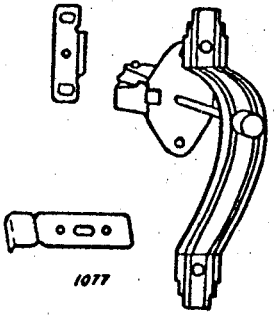
CABINET LATCHES:


1076

Type 1076.—Wrought brass handle and push button for outside of door; steel spring catch inside; all applied with screws from back. Die-pressed handle of modern design, pattern optional; grip with closed back, flat or rounded; over-all length 3⅜" to 5". Washer-head machine screws of steel for fastening handle. Catch with bevel or wing bolt, reversible for right- and left-hand flush and lipped doors. Plunger for doors up to 1⅛" thick. Steel parts cadmium-plated; exterior parts chromium-plated, unless otherwise specified.

Type 1076A.—Same as type 1076, except that handle may be of die-cast zinc alloy with plastic or nonferrous metal push button. Exterior metal parts chromium-plated.

Type F1076.—Same as type 1076, except all of steel; exterior parts chromium-plated.


1077

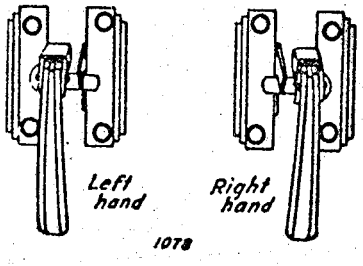
Type 1077.—Wrought brass handle and push button for outside of door; steel spring catch inside; applied with screws front

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CABINET LATCHES, CONT'D.:

and back. Handle 3½" to 4¼" x ½" to ⅝" x 0.040" minimum thickness. Catch with bevel or wing bolt reversible for right- and left-hand flush and lipped doors. Plunger for doors up to 1⅛" thick. Steel parts cadmium-plated; exterior parts chromium-plated, unless otherwise specified.

Type F1077.—Same as type 1077, except all of steel; exterior parts chromium-plated.

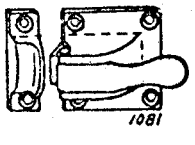


Type 1078.—Cast brass. Spring catch with pendant handle. For surface application to outside of ⅜" lipped doors, unless specified for flush doors. Right or left hand as required. Bolt contact either over or under strike, 2½" to 3½" x 1¼" to 1¾" over-all. Chromium-plated, unless otherwise specified.

Type 1078A.—Same as type 1078, except wrought brass, not less than 0.040" thick.

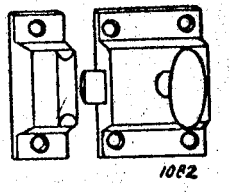
Type F1078A.—Same as type 1078, except wrought steel, not less than 0.040" thick, chromium-plated.

CABINET CATCH:



Type 1081.—Cast bronze. Size 1 7/16" to 1⅝" by 1 7/16" to 1½". Minimum throw of bolt ¼". Lever handle operated by outward pull.

CUPBOARD TURNS:



Type 1082.—Cast bronze. Oval knob. Size, without strike, approximately 1¾" x 1⅛". Latch bolt, minimum 11/32" x 5/16"; minimum throw 7/32".

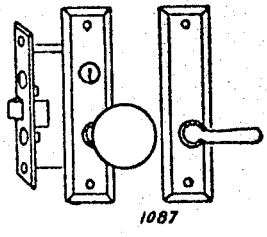
Type 1083.—Cast bronze. Oval knob. Size, without strike, approximately 2¼" x 1½". Latch bolt, minimum 15/32" x 7/16"; minimum throw 5/16".

Type 1085.—Wrought bronze. Minimum thickness of case 0.040". Otherwise the same as type 1083.

Type F1085.—Wrought steel. Minimum thickness of case 0.040". Otherwise the same as type 1083.

MORTISE SCREEN DOOR CATCHES:

CYLINDER TYPE:



Type 1087.

For 1⅛" to 1⅜" doors.

Case—Minimum 3" x 2¼" x ½". Cast iron.

Front—Minimum 4¼" x ¾". Bronze.

Bolt—Bronze.

Backset—1⅝".

Strike—Bronze.

Cylinder—Cast bronze. Three or more pin tumblers.

Keys—Two, white bronze. When specified, keys may be of aluminum alloy.

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MORTISE SCREEN DOOR CATCHES, CONT'D.:

CYLINDER TYPE, CONT'D.:

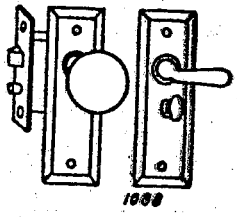
Knob—Minimum 1¾" diameter. Wrought bronze.

Lever handle—Minimum length 1¾". Cast bronze.

Escutcheons—Minimum 4½" x 1⅜". Wrought bronze. Beveled or rolled edges.

Operation: By knob or lever handle, except when outside knob is locked by stop; then by key from outside or lever handle inside. Reversible.

TURN KNOB TYPE:



Type 1088.

For 1⅛" to 1⅜" doors.

Case—Minimum 2" x 1¾" x ½". Cast iron.

Front—Minimum 3¼" x 1/16". Bronze.

Bolts—Bronze.

Backset—Approximately 1¼".

Strike—Bronze.

Knob—Minimum 1¾" diameter. Wrought bronze.

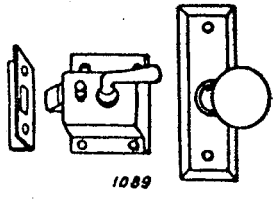
Lever handle—Minimum length 1½". Cast bronze.

Escutcheons — Minimum 4" x 1⅜". Wrought bronze. Beveled or rolled edges.

Turn knob—Cast bronze.

Operation: Latch bolt by knob or lever handle; dead bolt by turn knob inside. Reversible.

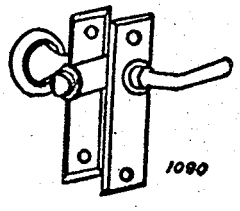
RIM SCREEN DOOR CATCH:



RIM SCREEN DOOR CATCH, CONT'D.:

Type 1089.—Cast or wrought bronze case, approximately 2¼" x 1⅞". Wrought bronze outside escutcheon with beveled or rolled edges, and knob; bronze lever. Stop in case to deadlock latch bolt. Bolt throw ½". For doors ⅞" to 1¼" thick, opening out.

TUBULAR SCREEN DOOR CATCH:



Type 1090.

Tubular case—Minimum ⅝" diameter x 2⅛" length. Brass or wrought steel.

Bolt—Brass. Minimum ⅝" diameter.

Backset—1¼".

Strike—Bronze, with lip.

Knob—Minimum 1¾" diameter. Wrought bronze.

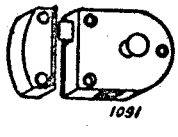
Lever handle—Minimum length 1⅝". Cast bronze.

Escutcheons — Minimum 4" x 1⅜". Wrought bronze. Beveled or rolled edges.

Stop—Bronze.

Operation: By knob or lever handle, except when bolt is deadlocked by stop on inside escutcheon. Reversible.

SECRET GATE LATCHES:

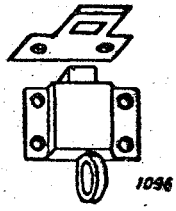


Type 1091.—Cast bronze. Single acting. Case 2" x 2⅜".

Type 1092.—Cast bronze. Double acting. Case 2" x 2⅜".

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TRANSOM CATCHES:


1096

Type 1096.—Cast bronze. Size 1¾" x 2¾". Latch bolt, minimum $\frac{21}{32}$" x ⅜". Minimum throw ½". Flat strike, unless otherwise specified.

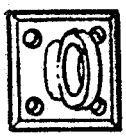
Type F1096.—Malleable iron. Otherwise the same as type 1096.

Type 1097.—Wrought bronze. Size 1¼" x 2". Latch bolt, minimum ½" x ⅜". Minimum throw $\frac{5}{16}$". Flat strike, unless otherwise specified.

Type F1097.—Wrought steel or cast iron. Otherwise the same as type 1097.

Type 1098.—Cast bronze. Size 1¼" x 2⅛", minimum. Independent acting latch bolt, minimum ½" x ⅜". Minimum throw $\frac{5}{16}$". Flat strike, unless otherwise specified.

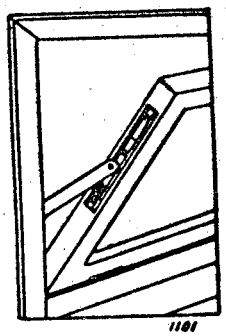
TRANSOM EYES:


1099

Type 1099.—One-piece cast bronze, 1¾" x 1¾". Projection 1½".

TRANSOM STAYS:

FRICTION, MORTISE TYPES:


1101

TRANSOM STAYS, CONT'D.:

FRICTION, MORTISE TYPES, CONT'D.:

Type 1101.—For light duty. Brass or bronze. Extruded channel ¾" x $\frac{11}{32}$" x $\frac{1}{16}$" minimum. Arm and bracket minimum ⅛" thick. Frictional element of asbestos non-abrasive brake lining having a minimum face area of $\frac{13}{16}$" square inch and held under adjustable pressure by metal spring. Free-acting shouldered pivots. Shock absorber in one end of channel. For maximum opening of 45°.

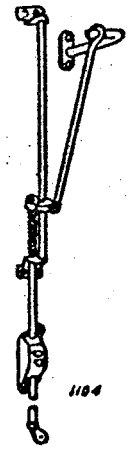
Type 1101A.—For heavy duty. Brass or bronze. Same as type 1101, except arm and bracket minimum $\frac{3}{16}$" thick. Extruded channel ⅝" x $\frac{11}{16}$" x $\frac{1}{16}$" minimum. Minimum effective face area of frictional element, 1½ square inches.

FRICTION, SURFACE TYPE:


1102

Type F1102.—Wrought steel, cadmium- or zinc-plated, with stop at 45° opening. Short arm not less than 4⅛" center to center. Long arm approximately 7".

TRANSOM LIFTERS:


1104

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TRANSOM LIFTERS, CONT'D.:

Type 1104.—Bronze, with spring. For bottom hung transoms. Automatic grip. Rod 3/8" diameter; 4 feet long, unless otherwise specified.

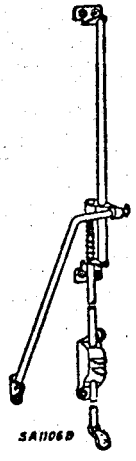
Type F1104.—Steel. Otherwise the same as type 1104.

Type F1105.—Steel. Rod 5/16" diameter. Otherwise the same as type 1104.

TRANSOM LIFTERS FOR SPECIAL APPLICATIONS (SEE 6.2):

Type SA1106A.—For center-hung transom opening in or out; for top-hung transom opening in. Bronze. Automatic grip. Rod 3/8" diameter; 4 feet long, unless otherwise specified. (Construction as for type 1104, except without spring.)

Type SA1106AF.—Steel. Otherwise the same as type SA1106A.



Type SA1106B.—For bottom-hung transom opening out. Bronze, with spring. Automatic grip. Rod 3/8" diameter; 4 feet long, unless otherwise specified.

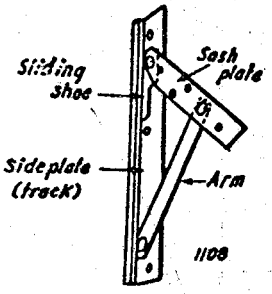
Type SA1106BF.—Steel. Otherwise the same as type SA1106B.

Type SA1106C.—Bronze; for top-hung transom opening out; without spring. Otherwise the same as type SA1106B.

Type SA1106CF.—Steel; for top-hung transom opening out; without spring. Otherwise the same as type SA1106B.

TRANSOM STAYS AND PIVOTS:

FRICTION TYPES:



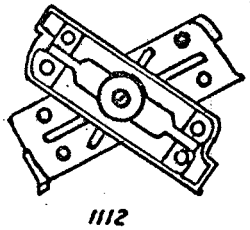
Type 1108.—Brass, unfinished. Applied in cut-out in sides of sash. Side plate, not less than 0.050" thick, with sliding shoe and friction spring. Sash plate not less than 3/32" thick. Arm not less than 1/8" thick. Lengths to suit sash (about 90 percent of height of sash). Shall hold sash at any opening up to horizontal. Pair, right and left, reversible for all applications.

Type F1108.—Steel, cadmium- or zinc-plated, with brass sliding shoe. Otherwise the same as type 1108.

Type 1108H.—Brass; same as type 1108, except for 1 3/4" heavy sash. Minimum thickness of sash plate 1/8", of arm 3/16". Side plate not less than 0.062" thick; finished width 1 3/8".

Type F1108H.—Steel, cadmium- or zinc-plated, with brass sliding shoe. For 1 3/4" heavy sash. Minimum thickness of sash plate 1/8", of arm 1/16". Side plate not less than 0.062" thick; finished width 1 3/8". Otherwise the same as type 1108.

SASH PIVOTS:



1112

Type 1112.—Cast bronze. Heavy coil friction spring. For 1/2" rabbet.

For 1 3/8" sash—Minimum size 4" x 1 3/8"
For 1 3/4" sash—Minimum size 4" x 1 3/4"

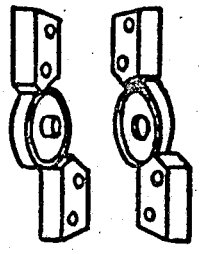
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SASH PIVOTS, CONT'D.:

For 2¼" sash—Minimum size 4½" x 2¼".

Type F1112.—Cast iron. Otherwise the same as type 1112.

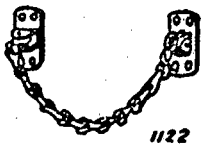


1114

Type 1114.—Cast bronze. Interlocking weatherproof grooves. For ½" rabbet.
For 1⅜" sash—size 3⅞" x 1⅜".
For 1¾" sash—size 3⅞" x 1¾".
For 2¼" sash—size 4" or 4½" x 2¼".

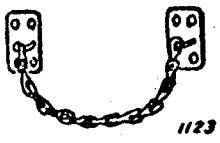
Type F1114.—Cast iron. Otherwise the same as type 1114.

TRANSOM CHAINS:



1122

Type 1122.—Heavy type. Bronze. Lugs and plate made in one casting. Four screws to each plate. Lengths 12" and 15". Bronze sash chain; minimum tensile strength 800 pounds.



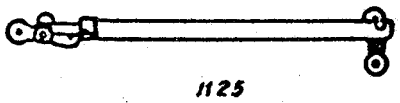
1123

Type 1123.—Wrought bronze. Four screws to each plate. Lengths 12" and 15". Bronze sash chain, minimum tensile strength 350 pounds.

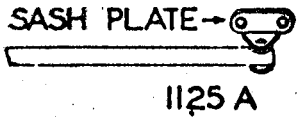
Type F1123.—Wrought steel. Otherwise the same as type 1123.

CASEMENT SASH ADJUSTERS:

TUBULAR FRICTION TYPES:



1125



1125 A

Type 1125.—Wrought bronze. Sliding rod and tube with friction adjusting nut knurled for finger hold. Minimum extension 6½". Lengths 10" and 12" for sash opening out; also suitable for use on outside of sash opening in.

Type 1125A.—For sash opening in, with adjuster mounted inside. Angle bracket for sash. Otherwise the same as type 1125.

BAR TYPES:



1126 SA1126

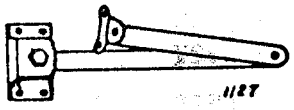
Type 1126.—Bronze. ⅜" diameter round rod. Swivel standard. Thumbscrew grip with broad knurled rim for finger hold. For sash opening out. Lengths 10" and 12".

Type F1126.—Cast iron or steel. Otherwise the same as type 1126.

Type SA1126.—(For special application, see 6.2.) For inswinging casements. Cast bronze. Rectangular bar $\frac{5}{16}$" by ½". Thumbscrew grip with broad knurled rim for finger hold. Lengths 12" and 15".

CASEMENT SASH STAYS:

FRICTION TYPE:



1127

Type F1127.—Wrought steel, cadmium- or zinc-plated. Short arm not less than 4" center-to-center. Long arm approximately

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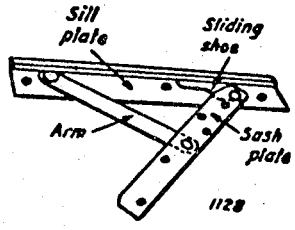
CASEMENT SASH STAYS, CONT'D.:

FRICTION TYPE, CONT'D.:

7" center-to-center. To open approximately 180°. Friction adjustment by spring washer and screw or bolt.

CASEMENT ADJUSTERS AND PIVOTS:

FRICTION TYPES:



Type 1128.—Brass, unfinished. For 1⅜" wood sash, swinging in or out. Applied in cut-out in top and bottom of sash. Sill plate, not less than 0.050" thick, and with sliding shoe and friction spring. Sash plate not less than $\frac{3}{32}$" thick. Arm not less than ⅛" thick. Lengths to suit sash (about ½ width of sash). Shall hold sash at any angle, and when wide open, inner edge of sash shall clear casing sufficiently to permit cleaning outside of sash. Pair, right and left, reversible.

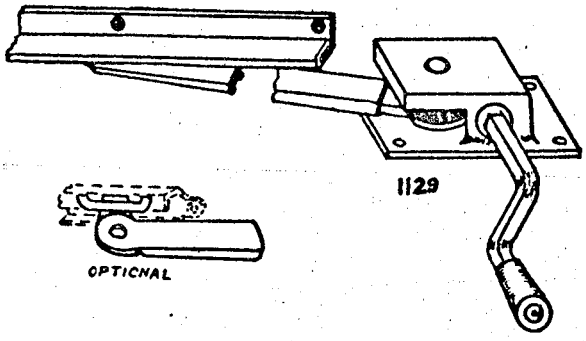
Type F1128.—Steel, cadmium- or zinc-plated, with brass sliding shoe. Otherwise the same as type 1128.

Type 1128H.—Brass; same as type 1128, except for 1¾" heavy sash. Minimum thickness of sash plate ⅛"; of arm $\frac{3}{16}$". Sill plate not less than 0.062" thick; finished width 1⅜".

Type F1128H.—Steel, cadmium- or zinc-plated, with brass sliding shoe. For 1¾" heavy sash. Minimum thickness of sash plate ⅛"; of arm $\frac{3}{16}$". Otherwise the same as type 1128. Sill plate not less than 0.062" thick; finished width 1⅜".

CASEMENT SASH OPERATORS:

HORIZONTAL TYPES:



Type 1129.—Enclosed worm and operating gear segment; arm with suitable sliding connection to sash. Worm of bronze or stainless steel; gear of hard bronze or monel metal; all other parts of brass. Length of gear along bottom of teeth, not less than ¼" measured parallel to axis. All parts and assemblies shall be of such strength that a gradually applied load of 175 pounds at end of arm, or at least 9½" from gear center, will not cause damage. Operation by brass crank handle with brass swivel knob grip. Unless otherwise specified, handle shall be detachable and have shaft of sufficient length to fit stools up to 4½" wide. Shall be furnished with supporting bracket for handle shaft when stools 6" or wider are specified.

Type F1129.—Same as type 1129, except that operating arm, sash slide track, and housing for worm and gear shall be of steel, cadmium- or zinc-plated (par. 3.9.1). Either worm or gear shall be bronze; if gear is steel, it shall have a brass pivot bushing. Operating arm shall have bronze spring or button to operate in the slide track. Handle of steel with brass swivel knob grip.

Type F1129A.—Same as type F1129, except that housing and spacing washers for gear may be zinc-alloy die castings. The die

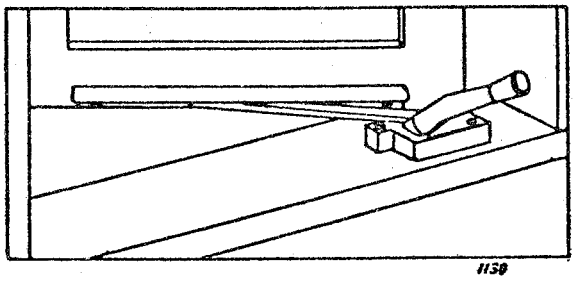
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CASEMENT SASH OPERATORS, CONT'D.:

HORIZONTAL TYPES, CONT'D.:

castings shall be phosphate-coated and lacquered with metallic finish lacquer on exposed surfaces.

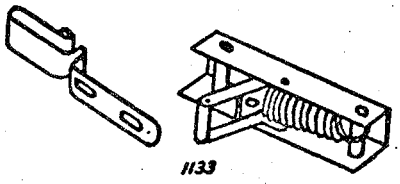
ANGLE DRIVE TYPES:



Type 1130.—Enclosed tilted worm and operating gear segment; arm and suitable sliding connection to sash. Worm of hard bronze or stainless steel; gear of hard bronze of suitable alloy to mate with worm, or monel metal; all other parts of brass or bronze. Length of gear along bottom of teeth, not less than $\frac{3}{16}$" measured parallel to axis. All parts and assemblies shall be of such strength that a gradually applied load of 125 pounds at end of arm, or at least 9½" from gear center, will not cause damage. Operation by brass or bronze crank handle with swivel knob grip. Handle shall be permanently attached.

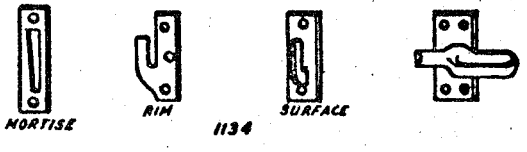
Type F1130.—Same as type 1130, except with steel or die-cast zinc-alloy housing and handle, steel operating arm, and steel slide track with bronze slide or button. Either worm or gear shall be bronze. If gear is steel, it shall have a brass pivot bushing. Length of gear along bottom of teeth, not less than $\frac{5}{32}$" measured parallel to axis. Steel parts cadmium- or zinc-plated (par 3.9.1); die castings shall be phosphate-coated and lacquered with metallic finish lacquer on exposed surfaces.

CASEMENT SASH TOP CLOSER:

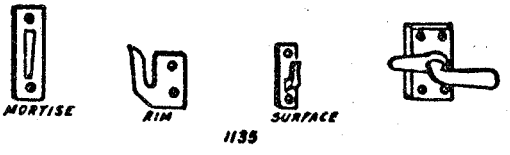


Type F1133.—Wrought steel, cadmium- or zinc-plated. Spring-actuated lever pivoted in formed case and arranged to engage hook sash bracket on sash when not less than 1" from closed position. On opening of sash, lever shall move to and remain in position for engagement on closing sash. Minimum width of case 1". Minimum length $3\frac{3}{16}$".

CASEMENT FASTENERS:

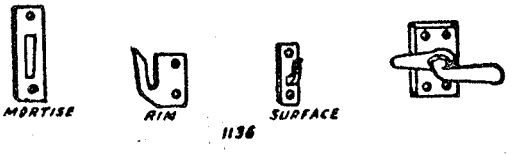


Type 1134.—Cast bronze. "D" handle on 3" x 1⅜" plate. Mortise, rim, or surface strike to suit application.



Type 1135.—Cast bronze. Lever handle on 2" x 1" minimum size plate. Projection 1½" minimum. Mortise, rim, or surface strike to suit application.

Type F1135.—Cast iron or wrought steel. Otherwise the same as type 1135.



Type 1136.—Cast or wrought bronze. Lever handle on 2" x 1" minimum size plate.

(Continued on next page)

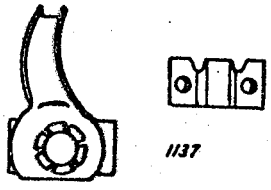
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CASEMENT FASTENERS, CONT'D.:

Projection 1". Mortise, rim, or surface strike to suit application.

Type F1136.—Cast iron or wrought steel. Otherwise the same as type 1136.

CELLAR WINDOW FASTENERS:



1137

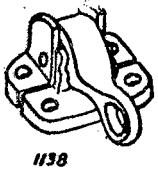
Type F1137.—Wrought steel, japanned finish. Cam action turn or strike; spring washer on turn-piece. Finger grip pull. Size over-all, 2½". Rim strike for sash opening in, unless otherwise required to suit application. [Commercial designs may differ widely from that illustrated.]

Type F1137C.—Wrought steel, cadmium- or zinc-plated (par. 3.9.1). Otherwise the same as type F1137.

SASH FASTENERS:

For double hung sash

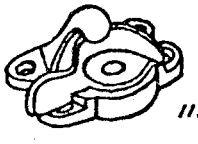
SIGNAL TYPE:



1138

Type 1138.—Cast bronze. Size of case, approximately 7/8" x $2\frac{5}{16}$". For high sash operated with window pole (type 1265A or 1265B).

CRESCENT TYPES:



1139

Type 1139.—Cast bronze. Heavy. Size of case, 1" x 2¾". Cam action.

SASH FASTENERS, CONT'D.:

CRESCENT TYPES, CONT'D.:

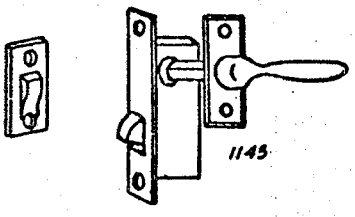
Type 1140.—Cast bronze. Light. Size of case, 7/8" x 2½". Cam action.

Type F1140.—Cast iron. Otherwise the same as type 1140.

Type 1142.—Wrought brass, either crescent or eccentric cam type. Size approximately 2" x 2".

Type F1142.—Wrought steel. Otherwise the same as type 1142.

GRILLE CATCH:



1145

Type 1145.—

Case—$2\frac{3}{8}$" x 7/8". Cast iron.

Front—3¼" x ¾". Cast bronze.

Backset—$\frac{7}{16}$".

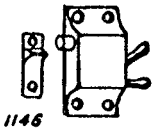
Strikes—Two; each 1⅝" x ¾" with $\frac{7}{32}$" minimum projection. Cast bronze.

Lever—2" length. Cast bronze.

Escutcheon—2" x ¾". Cast bronze.

Operation: Catch, mounted in jamb of opening, shall release grille on operation of handle; strike mounted on grille.

CAR WINDOW CATCH:



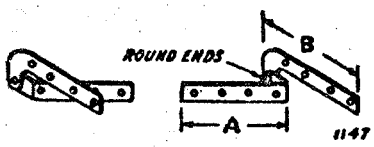
1146

Type 1146.—Cast bronze. Approximately 2½" by 1¼". Right and left hand. Furnish with three stops, about 1½" by $\frac{7}{16}$", with $\frac{7}{32}$" minimum projection.

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SCUTTLE HARDWARE:

SCUTTLE HINGES:

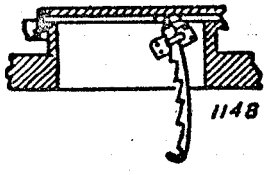


Type F1147.—Wrought steel, ¼" x 1½"; heavy zinc coating (par. 3.10.1). Spacing washer between leaves for free operation. Pair, right- and left-hand. Dimensions A and B as follows:

	Inches	*Inches*
Size	10	12
Dimension A	10	12
Dimension B	12	24

Type F1147C.—Wrought steel, cadmium- or zinc-plated (par. 3.9.1). Otherwise the same as type F1147.

SCUTTLE FASTENERS:

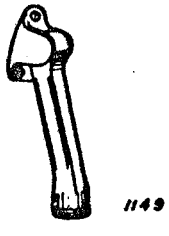


Type F1148.—Wrought steel; heavy zinc coating (par. 3.10.1). Size $\frac{1}{16}$" x 1½" x 30". Shall lock scuttle automatically in closed position.

Type F1148C.—Wrought steel, cadmium- or zinc-plated (par. 3.9.1). Otherwise the same as type F1148.

DOOR HOLDERS (nonautomatic):

LEVER TYPES:



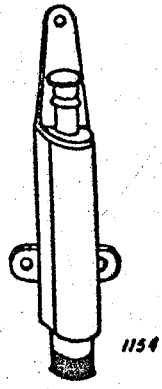
Type 1149A.—Cast bronze. Replaceable rubber tip. Lever shall swing up and remain securely held in vertical position when out of action. Length 4" to 4½" for doors with 2" or less floor clearance.

DOOR HOLDERS (nonautomatic), CONT'D.:

LEVER TYPES, CONT'D.:

Type 1149B.—Same as type 1149A, except with lever 5" to 5½" long for doors with 3½" or less floor clearance.

PLUNGER TYPES:



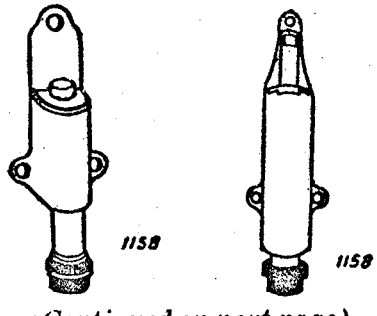
Type 1154.—Cast bronze case; steel catch plate and steel or malleable iron plunger rod with rubber tip. Steel parts plated. Rubber tip replaceable. Releasing lip shall be plainly evident. Size 7¼" (over-all height); minimum throw of plunger 1".

Type 1154A.—Cast bronze case. Same as type 1154A, except 6" size.

Type F1154A.—Cast iron case. Size 6". Otherwise the same as type 1154.

Type 1156.—Wrought bronze case. Size 6". Otherwise the same as type 1154.

Type F1156.—Wrought steel case. Size 6". Otherwise the same as type 1154.



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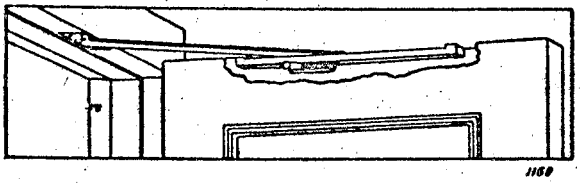
DOOR HOLDERS (nonautomatic), CONT'D.:

PLUNGER TYPES, CONT'D.:

Type 1158.—Cast bronze case; steel catch plate and steel plunger rod with rubber shoe. Steel parts plated. Rubber shoe flanged and held in place by knurled and threaded bronze collar so as to be replaceable without removing holder from door. Size not less than 6¾" by $1\frac{5}{16}$" over-all. (Two acceptable designs are illustrated.) Releasing lip shall be plainly evident.

AUTOMATIC DOOR HOLDERS:

CONCEALED OVERHEAD SLIDE:



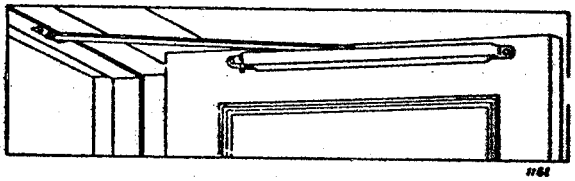
Type 1160.—For single-acting entrance and vestibule doors hung on butts and opening to 110°. Brass or bronze. Extruded channel 1¼" wide by ⅛" thick, minimum, fastening to door at each end and at not less than two intermediate points. Arm and bracket $\frac{3}{16}$" thick, minimum; widths to suit thickness of door, but not less than 1¼". Stop with encased shock absorber effective at all times. Hold-open and release by push and pull on door, except when exposed control knob is set in inactive position. Holding pressure adjustable for various conditions. Applied in mortise in top rail of door; arm exposed when door is open. Size to suit width, thickness, and swing of door.

Type 1160A.—Same as type 1160, except for doors hung on offset pivots.

Type 1160D.—For double-acting entrance and vestibule doors. Same as type 1160, except width of arm same as thickness of door.

AUTOMATIC DOOR HOLDERS, CONT'D.:

SURFACE-MOUNTED OVERHEAD SLIDE:



Type 1161.—For entrance and vestibule doors hung on butts and opening to 110°. Brass or bronze. Extruded track or channel. Arm and bracket ¼" thick, minimum. Stop with encased shock absorber effective at all times. Hold-open and release by push and pull on door, except when control knob is set in inactive position. Holding pressure adjustable for various conditions. Applied to surface of top rail of door with ¼" or larger machine screws and sleeve bolts through door. Size to suit width, thickness, and swing of door.

Type 1161A.—Same as type 1161, except for doors hung on offset pivots.

SURFACE-MOUNTED, WITH OVERHEAD JOINT:



Type 1162.—For entrance and vestibule doors opening to 115°. Cast phosphor bronze. Disk-form middle joint providing large horizontal bearing area, and having stop lugs. Encased shock absorber with compression spring. Hold-open and release by manual operation of holding catch. Stop and shock absorber effective at all times. Furnish 15" extension handle when specified.

Size A—For doors 24" to 29" wide. Minimum weight, 3 lbs.

Size B—For doors 30" to 35" wide. Minimum weight, 3 lbs., 10 oz.

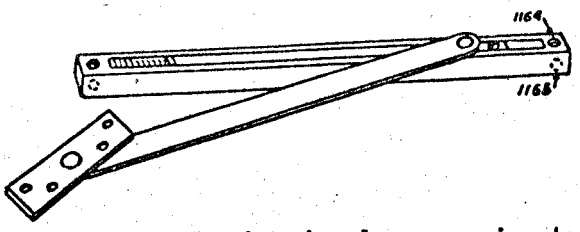
Size C—For doors over 35" wide. Minimum weight, 3 lbs., 14 oz.

Type F1162.—Same as type 1162, except with malleable iron arms, plated; bronze bearings and fittings.

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AUTOMATIC DOOR HOLDERS, CONT'D.:

CONCEALED OVERHEAD FRICTION SLIDE:

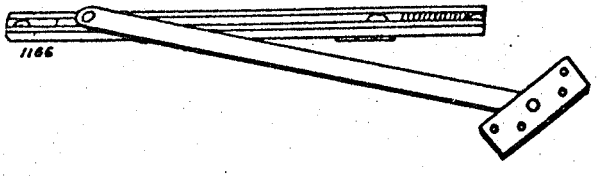


Type 1164.—For interior doors opening to 110°. Brass or bronze. Extruded channel 5/8" x 1 1/8" x 1/16" thick, minimum. Arm and bracket 3/16" thick, minimum. Frictional element made of asbestos nonabrasive brake lining having a minimum effective face area of 1¼ sq. in., and held under adjustable pressure by metal spring. Free-acting shouldered pivots. Shock absorber in one end of channel. Applied in mortise in top rail of door. Size to suit width, thickness, and swing of door. Shall hold in all positions.

SURFACE-MOUNTED OVERHEAD FRICTION SLIDE:

Type 1165.—For interior doors opening to 110°. Same as type 1164, except for application to surface of top rail of door by machine screws and sleeve bolts through door.

CONCEALED OVERHEAD SLIDE:



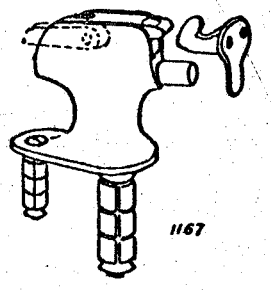
Type 1166.—For interior doors of medium size, single or double acting, opening to 110°. Bronze or steel channel and arm; channel 0.050" thick, minimum; arm and bracket 3/16" thick, minimum. Widths to suit thickness of door. Stop with shock absorber. Hold-open and release by push and pull on door. Applied in mortise in top rail of door; arm exposed when door is open. Size to suit width, thickness, and swing of door. Steel channel cadmium-plated.

AUTOMATIC DOOR HOLDERS, CONT'D.:

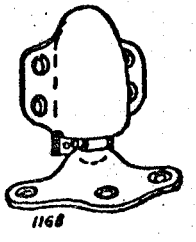
CONCEALED OVERHEAD SLIDE, CONT'D.:

Type 1166A.—Stay for interior doors of medium size. Same as type 1166, except without hold-open.

FLOOR TYPES:



Type 1167.—For entrance and vestibule doors. Cast bronze. Encased spring bumper with metal plunger and stop. Door strike plate with projecting hook to engage hinged holding tongue on top of bumper. Operation by push on door to the stop position, when engagement is automatic; release by quick push on door. Holding tongue inactive when placed in reverse position. Height 3½" for doors with 1½" maximum floor clearance, unless otherwise specified. Base at least 3½" by 2". Attachment to floor by a ½" steel stud with expansion sleeve and a ¼" - 20 machine screw with expansion sleeve.

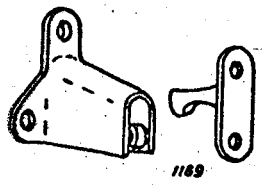


Type 1168.—Cast bronze. Push-and-pull. Spring-actuated engaging roller encased in housing attached to door by four screws. Strike attached to floor by three screws. Furnish screws for wood or metal, or machine screws and expansion sleeves for masonry, to suit required application. Rubber bumper. Housing at least 3¼" x 2" at base; 2" maximum projection. Strike approximately 2" x 4" at base with height to suit clearance between door and floor.

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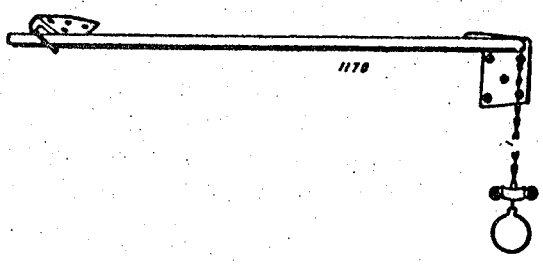
AUTOMATIC DOOR HOLDERS, CONT'D.:

WALL TYPE:



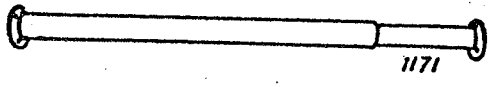
Type 1169.—Cast bronze. Push-and-pull. Spring-actuated engaging roller encased in housing applied to wall by not less than three screws. Strike applied to door by two screws. Furnish screws for wood or metal, or machine screws and expansion sleeves for masonry, to suit required application. Maximum projection of housing 2¼″; of strike 2⅛″. Clearance between door and wall when engaged, 3″ maximum, 2½″ minimum. Rubber bumper.

OVERHEAD GARAGE TYPE:



Type F1170.—Wrought steel, japanned finish. Bar not less than 30″ long and of U-shaped cross section. Catch on end of bar shall prevent pulling through bumper plate. Release chain approximately 38″ long. Reversible for right- or left-hand doors.

CLOSET HANGER BARS:



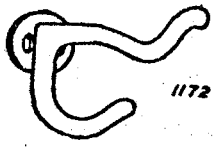
Type 1171.—Wrought brass telescoping tubing, nickel plated. Minimum wall thickness $\frac{1}{32}$″. Seamless outer tube not less than 1″ outside diameter. A metal wall plate shall be securely attached to each end of bar, and shall have at least two screw holes for attaching to hook strip in closet. Size 18″ adjustable from 18″ to 30″; size

CLOSET HANGER BARS, CONT'D.:

30″, adjustable from 30″ to 48″; and size 48″, adjustable from 48″ to 72″. Furnish center support with 48″ size.

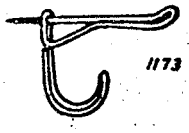
Type F1171.—Wrought steel, butted tube. Otherwise the same as type 1171.

COAT AND HAT HOOKS:



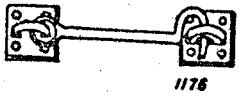
Type 1172.—Cast bronze. Upper prong bent. Two screws. Projection 3″. Commercial finish.

Type F1172.—Cast iron. Otherwise the same as type 1172.

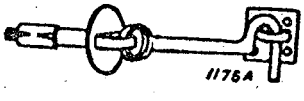


Type F1173.—Steel wire. Diameter 0.135″ (No. 10 gage). With protective coating. Screw point. Projection 3″.

DOOR HOOKS:



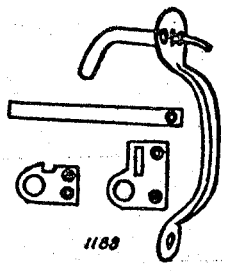
Type 1176.—Cast bronze. Lengths 3″, 4″, 5″, and 6″. Applied with wood screws. Commercial finish.



Type 1176A.—Cast bronze. Lengths 3″, 4″, 5″, and 6″. Hook applied with expansion bolt. Eye plate applied with wood screws. Commercial finish.

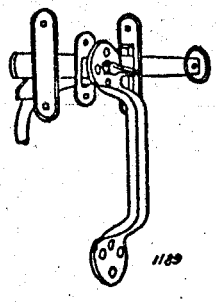
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THUMB LATCHES:

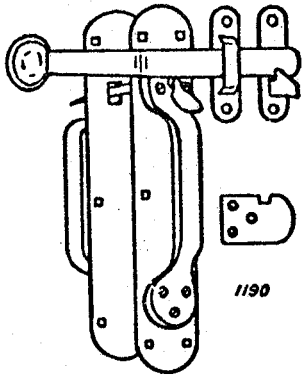


Type F1188.—Wrought steel, japanned finish. Strike and guide with padlock eyes.

	Inches	*Inches*
Sizes	7½	8¾
Length of handle	7½	8¾
Length of latch bar	5½	5½
For door thickness	¾ to 1⅞	1 to 2⅝



Type F1189.—Wrought steel, extra heavy; japanned finish. For doors 1¾″ to 2½″ thick. Handle approximately 9½″ over-all.

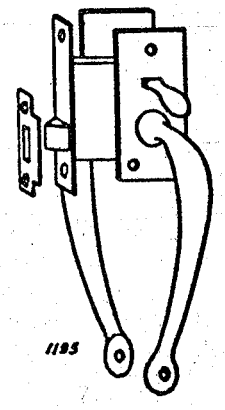


Type F1190.—Wrought steel, extra heavy; japanned finish. For doors 1¾″ to 2¼″ thick. Two handles; each approximately 9½″ over-all, attached to plates 14½″ in length.

THUMB LATCHES, CONT'D.:

Type F1190C.—Wrought steel, cadmium- or zinc-plated (par. 3.9.1). Otherwise the same as type F1190.

MORTISE THUMB LATCHES:



Type F1195.—For doors 1⅝″ to 2¼″ thick. Wrought steel, japanned.

Case—Minimum 3¼″ x 2⅝″ x ¾″.

Front—Minimum 4⅜″ x 1″.

Backset—2¼″.

Latch bolt—¾″ minimum throw.

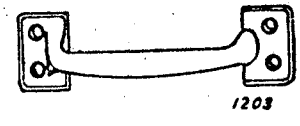
Handles—Two; each approximately 9¼″ over-all.

Operation: Latch bolt, by thumb piece from either side. Reversible.

Type F1195C.—Wrought steel, cadmium- or zinc-plated (par. 3.9.1). Otherwise the same as type F1195.

Type F1195H.—Wrought steel, heavy zinc coating (par. 3.10.1). Otherwise the same as type F1195.

BAR SASH LIFTS:



Type 1203.—Cast bronze. Heavy. Rounded corners. 1⅜″ x 5″ on bases.

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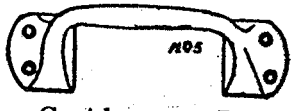
BAR SASH LIFTS, CONT'D.:

Type 1204.—Cast bronze. Light. Rounded corners. $1\frac{1}{16}$" x 4" on bases.

Type F1204.—Cast iron. Otherwise the same as type 1204.

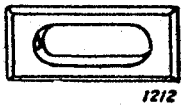
Type F1204A.—Wrought steel. Otherwise the same as type 1204.

OFFSET TYPE:



Type 1205.—Cast bronze. Rounded corners. 1¼" x 5" on bases.

FLUSH SASH LIFTS:

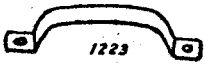


Type 1212.—Cast bronze. Beveled edge. Square corners. Size 1¾" x 3". Side screw holes. Lifting surfaces shall be smooth and free from sharp, rough edges.

Type 1213.—Wrought bronze. Otherwise the same as type 1212.

Type F1213.—Wrought steel. Otherwise the same as type 1212.

SCREEN LIFTS:

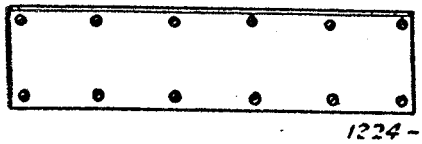


Type 1223.—Wrought brass. Length 3½".

Type F1223.—Wrought steel, japanned finish. Length 3½".

Type F1223C.—Wrought steel, cadmium- or zinc-plated (par. 3.9.1). Length 3½".

KICK PLATES, MOP PLATES, STRETCHER PLATES, AND ARMOR PLATES:



KICK PLATES, MOP PLATES, STRETCHER PLATES, AND ARMOR PLATES, CONT'D.:

Type 1224.—Wrought brass. Square corners, edges not beveled, unless beveled edges are specified. Metal 0.062" thick. Oval-head screws, spaced not over 8" on centers, unless otherwise specified. Height and width as specified.

Type 1225.—Stainless steel, minimum 18 percent chromium and 8 percent nickel. Commercial brushed finish similar to US26D. Square corners, edges not beveled, unless beveled edges are specified. Metal 0.050" thick. Oval-head screws, spaced not over 8" on centers, unless otherwise specified. Height and width as specified.

Type 1226.—Wrought aluminum, with anodic film (see 3.14). Square corners, edges not beveled, unless beveled edges are specified. Metal 0.062" thick. Commercial dull finish, similar to US26D. Oval-head screws, spaced not over 8" on centers, unless otherwise specified. Height and width as specified.

Type 1226A.—Same as type 1226, except with bright finish.

Type 1227.—Plastic plate ⅛" thick. Black color, unless chocolate brown is specified, with uniform satin matte finish on face. Core shall be of substantially the same color as the face. Square corners; edges beveled $\frac{1}{16}$" minimum at 45°. Oval-head brass screws finished to harmonize with plate. Screws spaced not over 8" on centers, unless otherwise specified. Height and width as specified.

(a) *Material.*—The plastic material shall be a laminate, molded in sheet form with suitable base or filler properly impregnated and bonded with a thermosetting resin compound or binder processed to make it hard, nonabsorbent and free from porosity and brittleness, so as not to chip or crack in normal service, and so as to meet the following requirements in addition to the requirements of 3.5.3, 3.5.4, and 3.5.5:

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KICK PLATES, MOP PLATES, STRETCHER PLATES, AND ARMOR PLATES, CONT'D.:

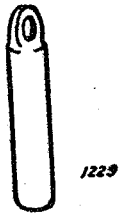
(b) *Impact strength.* — The plates shall have an impact strength of not less than 0.50 foot-pounds per inch of notch at 25°C. (77°F.) when tested as specified in 4.8.5.

(c) *Resistance to boiling water.*—The increase in weight after boiling in water for 2 hours shall not exceed 3 percent of the original weight, and the increase in thickness of the same samples shall not exceed 5 percent of the original thickness, when tested as specified in 4.8.6. Visual examination of samples after the boiling test shall reveal no delamination, cracking, or splitting and no blistering, crazing, or checking of the surfaces.

(d) *Abrasion resistance.*—In the abrasion test specified in 4.8.7, the average loss in weight shall not exceed 30 milligrams per 1000 revolutions.

(e) *Resistance to chemicals.*—The face of the plates shall show no spotting or blistering after 24 hours exposure to chemicals as specified in 4.8.8.

SASH WEIGHTS:[4]



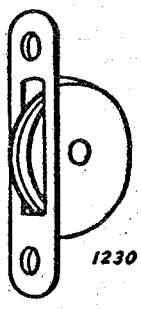
Type 1229F.—Cast iron. Round; reasonably smooth; free from fins; with well-formed eyes. Weights as required.

Type 1229FS.—Same as type 1229F except square.

Type 1229L.—Same as type 1229F except cast lead, round.

Type 1229LS.—Same as type 1229F except cast lead, square.

SASH PULLEYS:



Type 1230.—Turned brass wheel, with combination groove for cord or chain. Minimum width of groove 0.360". Front not less than 5" x 1⅛" x ⅛", with wrought bronze face plate. Cast iron case with round ends. Roller bearing with ¼" minimum diameter steel axle and rollers machined true. Wheel diameter 2¼", unless 2½" diameter wheel is specified.

Type 1230A.—Turned iron wheel. Otherwise the same as type 1230.

Type 1230B.—Same as type 1230, except with turned iron wheel and plain axle. If wood bushing is used, it shall be of hardwood impregnated so as to withstand the test in 4.9.

Type SA1231.—(For special application, see 6.2.) Overhead type sash pulley. Turned-face iron wheel. Wrought bronze front. Cast-iron case. Roller bearings, rollers machined true. Wheels fully housed to prevent chain from leaving wheel. Equipment shall include all necessary parts of suitable sizes for complete operation on double-hung counter-weighted sash in mullion windows without weights in the mullions.

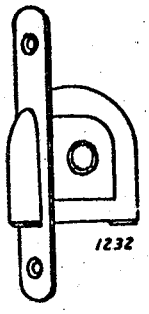
Type SA1231A.—Same as type SA1231, except with plain axle. If wood bushing is used, it shall be of hardwood impregnated so as to withstand the test in 4.9.

---

[4]See Federal Specification RR-C-271, Chain and Attachments; Standard, Miscellaneous, for sash chain and fasteners.

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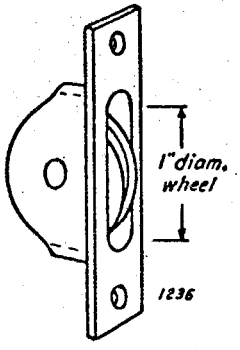
SASH PULLEYS, CONT'D.:



Type 1232.—Wrought bronze case and wheel. Enclosed type. Combination wheel with groove for cord or chain, 2½" in diameter. Front 5¾" x 1¼" x $\frac{3}{32}$". For frame stiles $1\frac{1}{16}$" or 1⅛" thick.

Type F1232.—Wrought steel case and wheel. Otherwise the same as type 1232. Shall be cadmium-coated before final finish.

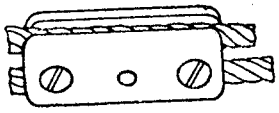
Type F1233.—Same as type F1232, except for 2" wheel. Front 4½" x 1" x $\frac{1}{16}$". For frame stiles ⅞" thick.



Type 1236.—For sliding sash or grille. Cast brass or bronze case and wheel; steel axle. 3¼" x ½" front; 1" diameter wheel grooved for ⅛" diameter wire cord. Wheel with steel bushing and antifriction bearings.



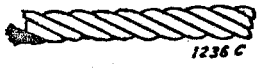
Soldering lug



Clamp

SASH PULLEYS, CONT'D.:

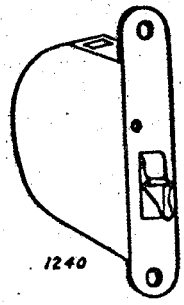
Type 1236A.—Cord attachments for use with type 1236 pulley. Lug for sash end, and clamp for weight end, both soldered. Shall be suitable for type 1236C wire cord.



Type 1236C.—Wire cord, for use with type 1236 pulley. Phosphor bronze. Six strands having 19 wires each. Cotton core. Overall diameter ⅛". Tensile strength at least 400 pounds. Length to suit.

SASH SPRING BALANCES:

MORTISE TYPES:



Type F1240.—Cast iron or steel face, plated (par. 3.6.2). Steel suspension tape, cadmium- or zinc-plated (par. 3.9.1). Cast iron or steel case; minimum thickness of metal 0.031". Plain axle. Uniformly tempered steel spring inclosed in cast iron or steel independent housing. Two balances per sash shall give complete counterbalance throughout entire travel of glazed sash. Side mounting unless overhead mounting is specified. Case and housing shall be cadmium- or zinc-plated (par. 3.9.1). Dimensions as tabulated on page 27.

Type 1240.—Wrought bronze face plate. Bronze suspension tape. Otherwise the same as type F1240.

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SASH SPRING BALANCES, CONT'D.:

MORTISE TYPES, CONT'D:

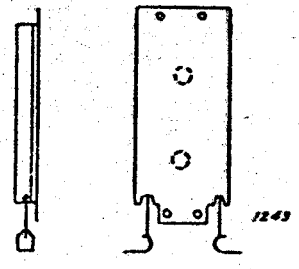
Dimensions for type F1240 and type 1240.

Maximum weight of glazed sash (see notes)	Minimum thickness of tape	Minimum tensile strength of tape	Minimum dimensions of face plate					
			Length		Width	Thickness		Minimum diameter of axle
			Overhead mounting	Side mounting		Wrought	Cast	
Pounds	*Inch*	*Pounds*	*Inches*	*Inches*	*Inches*	*Inch*	*Inch*	*Inch*
26	0.010	150	4¼	5½	¾	0.050	0.09	½
35	.010	150	4⅛	5½	⅞	.062	.12	⅝
56	.012	250	5⅞	6¾	1⅜	.062	.12	⅝
105	.016	600	7¼	8¾	1¾	.062	.12	¾

NOTE 1.—In supplying sash balances, the weight of each sash and its vertical travel are required. If weight is not available, the size and thickness of sash, material, kind of glass, and any other details necessary for estimating weight, should be indicated.

NOTE 2.—When specified in the invitation for bids, vertical balances, type F1245, may be furnished in lieu of type F1240 balances for sash weighing not over 80 pounds, provided that for sash weighing over 30 pounds, the vertical balances are of heavy construction for application in groove ½" wide by 1⅛" deep in stile of sash.

DUAL TYPES:



(For mounting flatwise in cut-out in side frame of window.)

Type F1243.—Steel mounting plate (face plate); two flat coil steel springs (clock type); two round steel suspension cables with clips for attaching to sash. Springs either adjustable or nonadjustable, enclosed in single case 6⅜" x 3¼" x ¾", or in two revolving drums, and adequately lubricated. Mounting plate not less than 0.031" thick; cable 0.050" minimum diameter, with tensile strength not less than 200 pounds. Plain axle not less than ½" diameter. Cadmium or zinc plating (par. 3.9.1) on cables and on all exterior surfaces, except that back cover plates or case may be lacquered. Two balances per window (two sash) shall give complete counterbalance throughout entire travel of each glazed sash. (Note.—The weight of each sash and its vertical travel are required.)

SASH SPRING BALANCES, CONT'D.:

DUAL TYPES, CONT'D.:

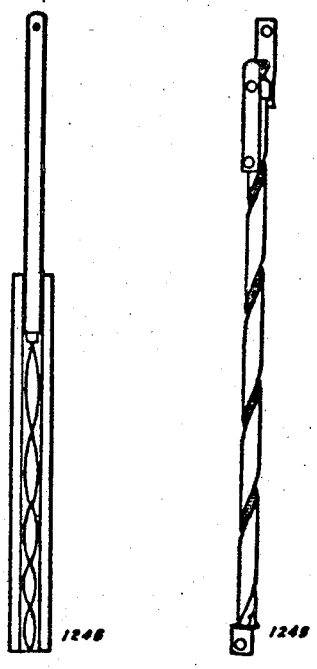
Type F1243A.—Same as type F1243, except one balance per window (two sash). For each sash, furnish one suitable spring equalizer or glide, for installation on side of sash opposite to the balance, to prevent jamming of sash. For sash not over 1⅜" thick and 36" wide, or 16 pounds maximum. For heavier sash, use type F1243.

VERTICAL TYPE:

Type F1245.—Vertical type for application in groove not over ⅝" wide by ⅞" deep in stile of sash. Spiral-actuated helical coiled spring completely concealed either in tubular housing or by stile of sash. Bearing surfaces shall be adequately lubricated at factory. Pair of balances shall fully counterbalance glazed sash at all points of travel and shall be furnished complete for attachment to sash weighing up to 30

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SASH SPRING BALANCES, CONT'D.:
VERTICAL TYPE, CONT'D.:



pounds. Exposed iron or steel parts cadmium- or zinc-plated (par. 3.9.1). (Note.—The weight of each sash and its vertical travel are required.) (Two of various optional constructions are illustrated.)

SASH POLE HANGERS:



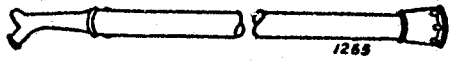
Type 1262.—Cast bronze. Size 2⅜" x 1⅛".

SASH POLE HOOKS:



Type 1264.—Cast bronze. Length 4⅛". Diameter 1⅛".

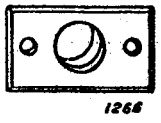
SASH POLES:



Type 1265A.—Bronze hook on handle of japanned or enameled steel tubing. Minimum wall thickness of tubing 0.035"; diameter ¾" ± $\frac{1}{16}$". Lower end of handle protected with rubber cap having steel washer insert. Length 6 feet, unless 8- or 10-foot lengths are specified.

Type 1265B.—Hardwood. 1⅛" minimum diameter. Chucked to fit sash pole hook. Length 6 feet long, unless otherwise specified. Poles exceeding 12 feet in length may be made in two sections with brass screw joint coupling.

SASH PULL PLATES:



Type 1266.—Cast bronze. Minimum size 1" x 2".

DOOR PULLS:[5]

LIGHT PULL

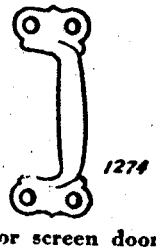


For screen doors:

Type 1269.—Wrought bronze. Rounded grip closed in back. Length 4¾".

Type F1269.—Wrought steel. Otherwise the same as type 1269.

HEAVY PULL



For screen doors:

Type 1274.—Cast bronze. Length 6". Four screw holes. Grip of flat oval section, with

[5]For door pulls classified with door trim, see Federal Specification FF–H–106.

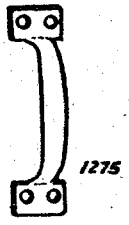
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DOOR PULLS, CONT'D.:

HEAVY PULL (For screen doors) CONT'D.:

all exposed surfaces polished. Minimum weight 4¼ lbs. per dozen without screws.

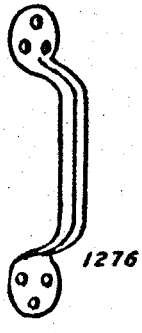
Type F1274.—Cast iron; japanned finish, unless otherwise specified. Otherwise the same as type 1274.



Type 1275.—Wrought bronze. Minimum length 6″. Flat oval grip closed in back. Four screw holes.

Type F1275.—Wrought steel, japanned finish. Minimum length 6″. Flat oval grip closed in back. Four screw holes.

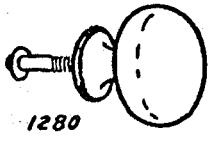
Type F1275C.—Wrought steel, cadmium- or zinc-plated (par. 3.9.1). Otherwise the same as type F1275.

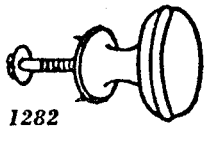


Type F1276.—Wrought steel, extra heavy. Japanned finish. Length approximately 9½″ over-all. Six screw holes.

Type F1276C.—Wrought steel, cadmium- or zinc-plated (par. 3.9.1). Otherwise the same as type F1276.

KNOB PULLS:





KNOB PULLS, CONT'D.:

Type 1280.—Cast bronze round knob and shank. Projections on base to prevent turning, unless otherwise required. Attachment by steel washer-head machine screw, cadmium-plated.

Sizes, inches	1	1⅛	1¼	1½
Diameter of knob, inches	1	1⅛	1¼	1½
Projection, min., inches	1	1	1⅛	1¼

Type 1280D.—Same as type 1280 except of zinc die-cast alloy. Chromium-plated.

Type 1280P.—Same as type 1280, except of plastic. Black color, unless otherwise specified.

Type 1281.—Seamless wrought bronze top; solid bronze shank. Otherwise the same as type 1280.

Type 1282.—Wrought brass knob and shank. Pattern optional. Otherwise the same as type 1280.

Type 1283.—Stainless steel cap; minimum 18 percent chromium, 8 percent nickel; commercial brushed finish similar to US26D. Wrought steel shank, chromium-plated. Pattern optional. Otherwise same as type 1280.

Type F1284.—Wrought steel knob and shank, chromium-plated. Pattern optional. Otherwise the same as type 1280.

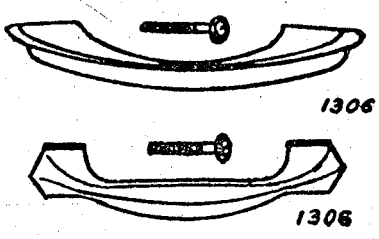
DRAWER PULLS:



Type 1305.—Cast bronze. Rounded handle; rectangular bases. Size, 4″ in over-all length. Two steel washer-head machine screws, cadmium-plated, for attachment from inside of drawer.

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DRAWER PULLS, CONT'D.:



Type 1306.—Cast bronze. Modern design, pattern optional. Size 4″ to 5″ in over-all length. Two steel washer-head machine screws, cadmium-plated, for attachment from inside of drawer.

Type 1306D.—Same as type 1306 except of die-cast zinc alloy, chromium-plated.

Type 1306P.—Same as type 1306, except of plastic. Black color, unless otherwise specified.

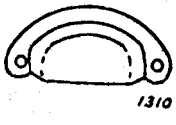
Type 1307.—Wrought brass; die pressed. Grip with closed back, flat or rounded. Size, 3⅜″ to 4¾″ in over-all length. Otherwise the same as type 1306.

Type F1307.—Wrought steel; die pressed. Chromium-plated. Grip with flat back. Size, 3⅜″ to 4¾″ in over-all length. Otherwise the same as type 1306.



Type 1308.—Wrought brass, minimum thickness 0.040″. Size 3½″ to 4½″ x ½″ to ⅝″ over-all. Two oval-head wood screws for attachment from front.

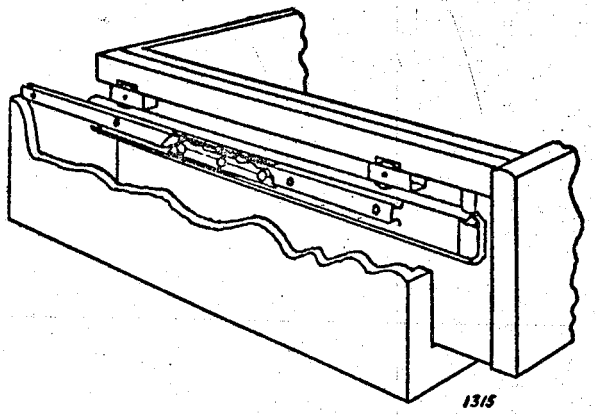
Type F1308.—Wrought steel, chromium-plated. Otherwise the same as type 1308.



Type 1310.—Wrought bronze. Cup pattern. Minimum length 3¼″.

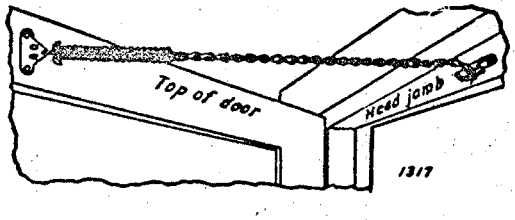
Type F1310.—Wrought steel, chromium-plated. Cup pattern. Length not less than 3¼″.

EXTENSION DRAWER SLIDE:



Type F1315.—For attachment to each side of drawer; maximum clearance each side between drawer and case, ¾″. Cold-rolled steel carriers not less than 0.080″ thick; extension member not less than 1¼″ by ⅝″ in cross section, carried on $\frac{5}{16}$″ diameter steel balls in stationary inner member, which is fastened to case. Drawer mounted on rollers which travel on upper edge of extension member. Rollers case hardened, not less than ⅞″ in diameter, and with ball bearings, roller bearings, or "oilite"-type bronze bearings, adequately lubricated. Carrier assembly reversible for either side of drawer. Resilient bumper stops at all striking points. All metal surfaces cadmium- or zinc-plated. Shall permit drawers to be pulled full length out of cabinet to stop position, and be lifted off carriers for removal from cabinet, if desired.

CHAIN DOOR STOP:



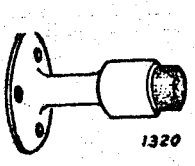
Type F1317.—For screen and storm doors opening out. Steel chain of flat link type in one length; compression spring or springs at least 4¼″ in total length and

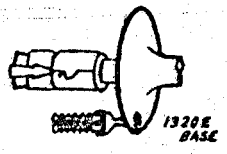
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CHAIN DOOR STOP, CONT'D.:

¾" in outside diameter attached to one or both brackets and chain. Spring shall withstand at least 100 pounds pull on chain before reaching full compression. Tensile strength of chain and attachments not less than 350 pounds. Brackets not less than 0.090" thick. Over-all length 22". Cadmium- or zinc-plated.

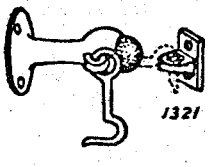
DOOR STOPS:

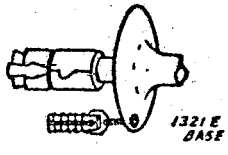




Type 1320.—Cast bronze, heavy weight. Rubber tip. Projection 3½". Diameter of base approximately 2¼". Three wood screws.

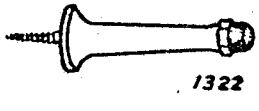
Type 1320E.—Same as type 1320, except for attachment by steel stud with expansion shield and auxiliary machine screw with expansion shield.





Type 1321.—Same as type 1320, except with hook and keeper.

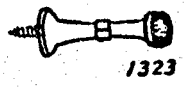
Type 1321E.—Same as type 1320, except with hook and keeper, and attachment by steel stud, with expansion shield and auxiliary machine screw with expansion shield.



Type 1322.—Cast bronze, medium weight. Rubber tip. Base with wood screw. Projection 3". Diameter of base approximately 1".

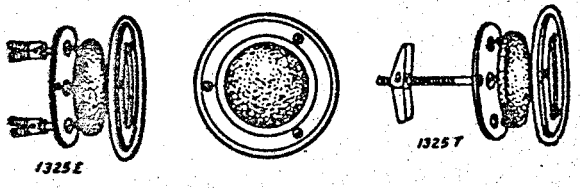
DOOR STOPS, CONT'D.:

Type F1322.—Cast iron. Otherwise the same as type 1322.



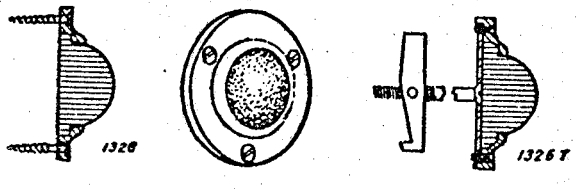
Type 1323.—Cast bronze, light weight. Rubber tip. Base with wood screw. Projection 2½". Diameter of base approximately 1".

Type F1323.—Cast iron. Otherwise the same as type 1323.



Type 1325E.—Cast bronze. Diameter 4". Convex rubber pad inserted from back into flanged rosette. Rosette secured to steel backplate attached to wall by expansion bolts.

Type 1325T.—Attachment to wall by toggle bolt. Otherwise the same as type 1325E.

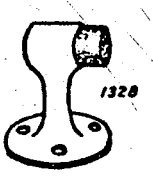


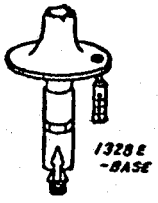
Type 1326.—Cast bronze. Diameter minimum 2¼". Convex rubber pad inserted from back into flanged rosette and attached to wall by three wood screws.

Type 1326T.—Same as type 1326, except backplate attached to wall by toggle bolt. Rosette secured to backplate by three machine screws.

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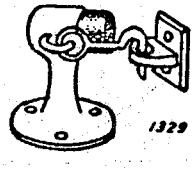
FLOOR STOPS:

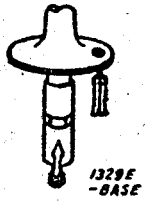




Type 1328.—Cast bronze, heavy weight. Rubber tip. Height approximately 3". Diameter of base approximately 2½". Attachment to floor by three wood screws.

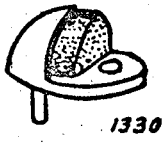
Type 1328E.—Same as type 1328, except for attachment to floor by steel stud with expansion shield and auxiliary machine screw with expansion shield.





Type 1329.—Same as type 1328, except with hook and keeper.

Type 1329E.—Same as type 1328, except with hook and keeper, and attachment to floor by steel stud with expansion shield and auxiliary machine screw with expansion shield.



Type 1330.—For doors without threshold. Cast bronze. Dome portion substantially quarter spherical. Molded rubber cushion fitting cavity in dome. Cushion approximately 1½" wide, ¾" high, and extending ¼" minimum beyond vertical face of dome. Cushion secured within dome by concealed fastenings. Attached to wood floor by No. 14 wood screw, unless ¼"—20 machine screw with expansion shield is required for attachment to other materials. Rear portion of base shall have a ¼" minimum diameter stud not less than ½" long, projecting into floor to prevent rotation. Minimum thickness of base $\frac{5}{32}$".

FLOOR STOPS, CONT'D.:

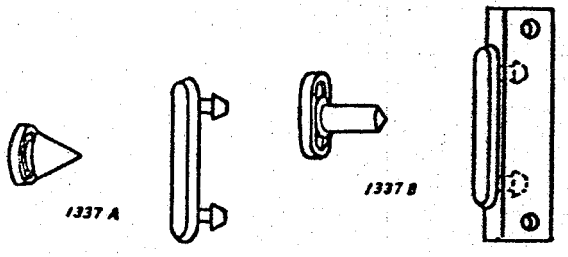
Type 1330A.—For doors with threshold. Minimum thickness of base ½". Otherwise the same as type 1330.



Type 1331.—Cast bronze. Rubber tip. Diameter of base 1¼". Height $1\frac{5}{16}$". Two wood screws.

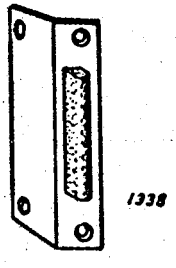
Type F1331.—Cast iron. Otherwise the same as type 1331.

DOOR SILENCERS:



Type 1337A.—For metal frame. Molded rubber with underside of head recessed to form pneumatic chamber to cushion impact of door. Diameter or width ⅜".

Type 1337B.—For wood frames. Otherwise the same as type 1337A. Bronze plate, if plate is used.

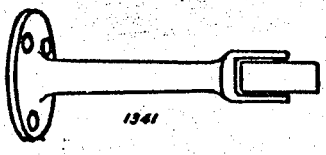


Type 1338.—For silencing doors in hospitals or institutions not equipped with door closers. Nonferrous metal case, approximately 2⅞" x $\frac{1}{16}$" x $\frac{9}{16}$". Partially encased rubber bumper backed by spring action.

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DOOR ROLLER-BUMPERS:

(For protecting interfering doors against striking each other.)


1341

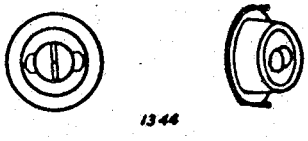

1342

Type 1341.—Cast bronze, with rubber roller vulcanized to a bronze bushing. Base approximately 2″ in diameter. Projection 4½″. Black roller, unless white is specified, not less than 1″ diameter by ⅜″ thick.

Type 1341A.—Same as type 1341, except projection 6½″.

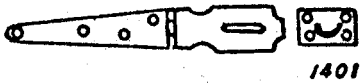
Type 1342.—Cast bronze with rubber roller vulcanized to a bronze bushing. Base approximately 2¼″ x 1½″. Projection beyond surface of door, 2¾″; overhang beyond edge of door, 3″. Black roller, unless white is specified, not less than 1″ diameter by ⅜″ thick.

WINDOW STOP ADJUSTER:


1344

Type 1344.—Wrought bronze cup. Adjustable. Number 8 round-head bronze screw, 1″ long.

HINGE HASPS:


1401

Type F1401.—Wrought steel; heavy zinc coating (par. 3.10.1). Brass pin. Dimensions as follows:

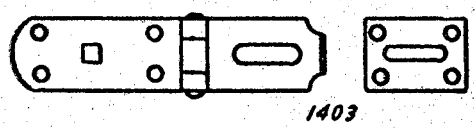
HINGE HASPS, CONT'D.:

Dimensions for type F1401

Length closed	Size of screws		Minimum thickness of metal	
	Length	No.	Hasp	Staple plate
Inches	*Inch*		*Inch*	*Inch*
3	¾	6	0.062	0.086
4½	¾	6	.074	.086
6	1	9	.084	.101
8	1	10	.096	.120

Type F1401C.—Wrought steel, cadmium- or zinc-plated (par. 3.9.1). Brass or steel pin. Otherwise the same as type F1401.

Type F1401E.—Wrought steel, unfinished. Steel pin. Otherwise the same as type F1401.

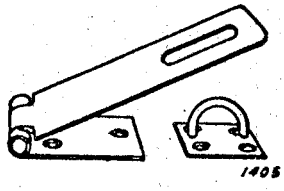

1403

Type F1403.—Extra heavy. Wrought steel; heavy zinc coating (par. 3.10.1). Brass pin. Length open, 7½″; width, 1¾″; thickness, 0.160″ minimum. One square hole shall be provided, for fastening by carriage bolt if desired. (Carriage bolt is not to be furnished.)

Type F1403C.—Wrought steel, cadmium- or zinc-plated (par. 3.9.1). Brass or steel pin. Otherwise the same as type F1403.

Type F1403E.—Wrought steel, unfinished. Steel pin. Otherwise the same as type F1403.

SAFETY PATTERN:


1405

Type F1405.—Wrought steel; heavy zinc coating (par. 3.10.1). Brass pin.. Dimensions as follows:

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HINGE HASPS, CONT'D.:

SAFETY PATTERN, CONT'D.:

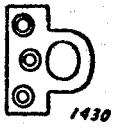
Dimensions for type F1405

Length closed	Length of joint	Size of screws		Number of screw holes	Minimum thickness of metal	
		Length	No.		Hasp	Staple plate
Inches	*Inches*	*Inches*			*Inch*	*Inch*
3½	1½	¾	7	7	0.074	0.062
4½	1½	⅞	7	7	.086	.062
6	1⅞	1	9	9	.104	.068
7	2½	1¼	10	9	.156	.156

Type F1405C.—Wrought steel, cadmium- or zinc-plated (par. 3.9.1). Brass or steel pin. Otherwise the same as type F1405.

Type F1405E.—Wrought steel, unfinished. Steel pin. Otherwise the same as type F1405.

PADLOCK EYES:



Type F1430.—Wrought steel; heavy zinc coating (par. 3.10.1). Size of plate approximately 2⅛" x 1⅞". Minimum thickness 0.104". Size of hole approximately ¾" x ⅝".

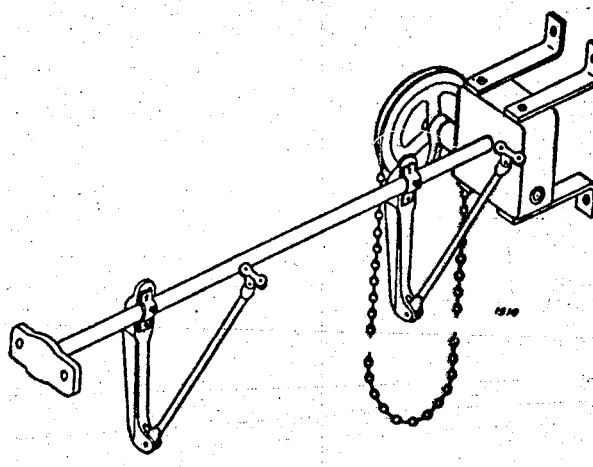
Type F1430C.—Wrought steel, cadmium- or zinc-plated (par. 3.9.1). Otherwise the same as type F1430.

Type F1430D.—Wrought steel, japanned. Otherwise the same as type F1430C.

SCUTTLE OPERATORS:

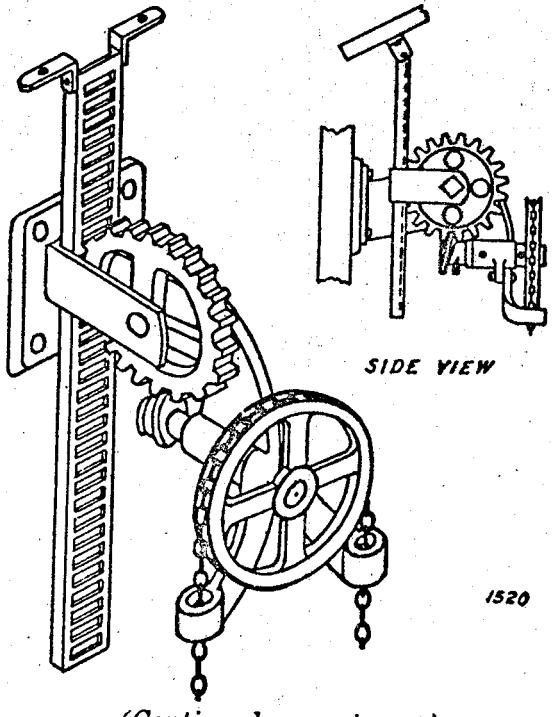
Type F1510.—For heavy scuttles. Operation by worm and gear with chain wheel and endless chain. Iron or steel cut gear with steel worm. Gear and worm enclosed in an oil-tight box with oil seals on all spindle openings. Shaft of 1" iron pipe pinned to the worm gear spindle. Worm

SCUTTLE OPERATORS, CONT'D.:



gear provided with proper stops to regulate the travel of gear. Arms and connections to the scuttle shall be of cast iron or wrought steel. Cast iron shaft bearing with machined bearing surface and oil hole. Arms secured to the shaft with clamp and set screw. Fastening to scuttle by wood screws. Fastening to curb by through bolts. Painted finish. Length of shaft and chain as specified.

SKYLIGHT OPERATORS:

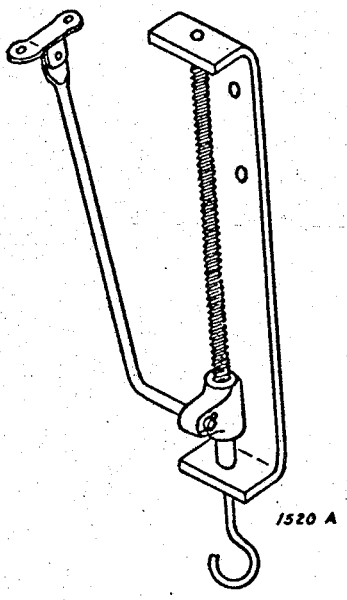


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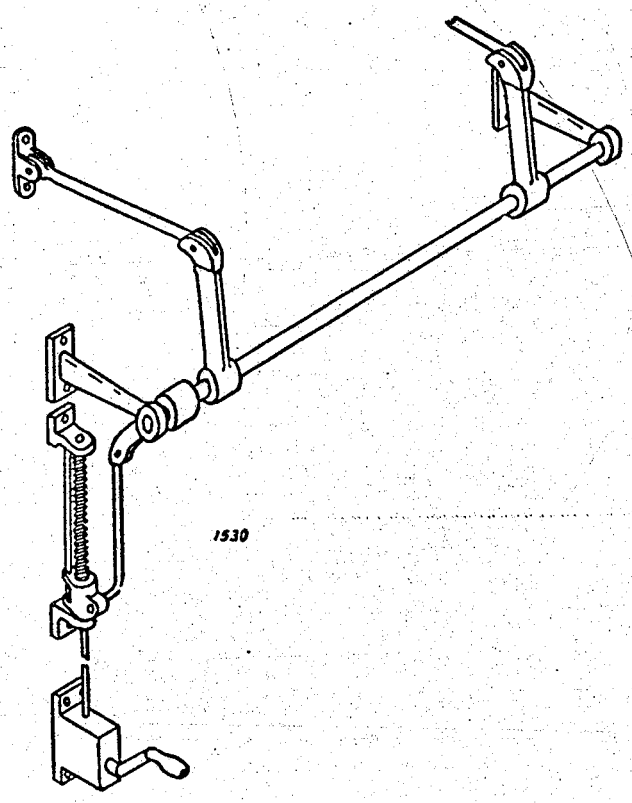
SKYLIGHT OPERATORS, CONT'D.:

Type F1520.—For skylights up to 5′ x 6′, or 30 square feet maximum area. Steel rack operated by cast-iron or cut-steel gear and worm with chain wheel and endless chain. Chain wheel keyed to shaft. Worm spindle equipped with a thrust bearing between the worm and the frame. Rack made of steel channel 1¼″ x ½″ x ⅛″ held in position by a wrought steel housing. Lift 20″ with 5″ extension of rack above center of gear. Painted finish.



Type F1520A.—For skylights up to 2′6″ x 3′6″ or 8¾ square feet maximum area. Steel lifting rod operated by a bronze nut traveling on a steel screw, and operated by a detachable pole and eye. Lifting rod $\frac{5}{16}$″ diameter. Frame for screw of $\frac{3}{16}$″ x 2″ steel. Screw $\frac{13}{32}$″ diameter. Lift 10″. Cadmium- or zinc-plated finish. Furnish without operating pole unless pole is specified.

TRANSOM OR SASH OPERATORS:

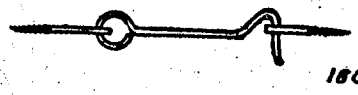


Type F1530.—For heavy sash over 3½′ wide, but not over 4′ x 5′. Horizontal steel rod ¾″ in diameter, supported by cast iron brackets with machined bearings. Rod located near the bottom of sash and connected thereto at each end by cast iron levers and steel connecting rods with cast iron or wrought steel sash plate. Horizontal shaft controlled by a vertical brass screw ½″ in diameter with cast iron traveling nut. Nut connected to an operating lever by a steel connecting rod. Operation by hand crank through enclosed cut steel miter gears. Cast iron sash levers and operating lever secured to shaft with set screws. Self-locking in all positions. Cadmium- or zinc-plated finish.

Type F1530A.—Same as type F1530, except with ⅞″ diameter horizontal rod and ¾″ diameter brass screw.

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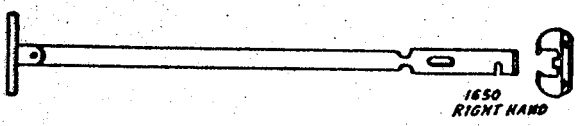
HOOKS AND EYES:



Type 1601.—Brass wire hooks and eyes. Hook lengths 1, 2, 3, 4, and 6 inches.

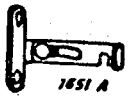
Type F1601.—Bright steel wire hooks and eyes. Hook lengths 1, 2, 3, 4, and 6 inches. Not plated.

STORM SASH AND SCREEN FASTENERS:



Type F1650.—Wrought steel, cadmium- or zinc-plated (par. 3.9.1). Length 10″ for sash. Shall hold sash open and closed. One pair, right and left hand, for each sash.

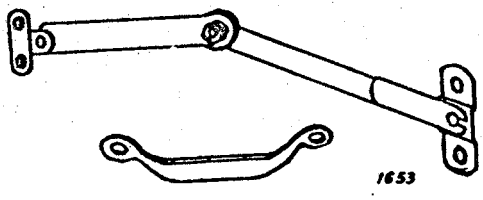
Type F1651.—Wrought steel, cadmium- or zinc-plated (par. 3.9.1). Length 3″ maximum for screen. Shall hold screen open and closed, and lock in closed position. One pair, right and left, for each screen.



Type F1651A.—Same as type F1651, except without catch; shall operate with catch furnished with type F1650.

STORM SASH FASTENER SETS:

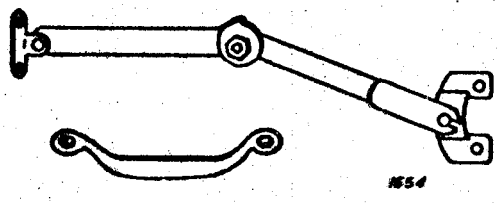
FRICTION TYPES:



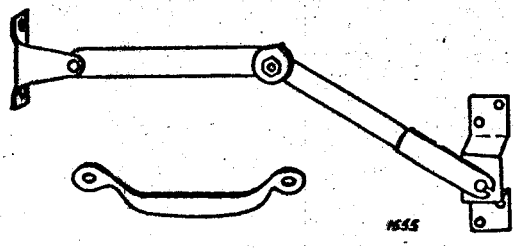
Type F1653.—Wrought steel, cadmium- or zinc-plated (par. 3.9.1). Two hinged arms with adjustable friction joint and suitable attaching brackets permitting removal of sash without removing brackets. Shall hold sash open, closed, and in any position between. Approximate length, 9″. One pair of fasteners, right and left hand, and one pull for each sash, unless otherwise specified.

STORM SASH FASTENER SETS, CONT'D.:

FRICTION TYPES, CONT'D.:



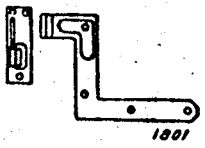
Type F1654.—For rabbeted sash. Otherwise the same as type F1653.



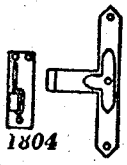
Type F1655.—For windows equipped with sliding screens. Otherwise the same as type F1653.

BLIND HARDWARE:

HINGES FOR BLINDS:



TYPE F1801, TYPE 1805
Left hand when used at bottom;
Right hand when used at top



TYPE F1804
TYPE F1806

Type F1801.—For blinds hung flush with the casing; frame construction. Wrought steel; heavy zinc coating (par. 3.10.1). Corrugated elbow. Hinge 4″ vertical by 5″ horizontal. Width 1″. Hook plate 3⅛″ x 1⅛″. Offset 1″. Throw 2″. A pair of blinds requires four hinges, two right hand and two left hand.

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BLIND HARDWARE, CONT'D.:

HINGES FOR BLINDS, CONT'D.:

Type F1801C.—Same as type F1801, except cadmium- or zinc-plated (par. 3.9.1).

Type F1804.—"T" form center pattern for frame construction. Length 6"; width 7/8". One pair required for each pair of blinds. Reversible for either hand. Otherwise the same as type F1801 with heavy zinc coating.

Type F1804C.—Same as type F1804, except cadmium- or zinc-plated (par. 3.9.1).

Type F1805.—For brick construction. Offset 2"; throw 4". Otherwise the same as type F1801 with heavy zinc coating.

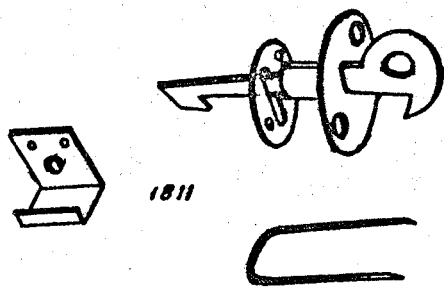
Type F1805C.—Same as type F1805, except cadmium- or zinc-plated (par. 3.9.1).

Type F1806.—"T" form center pattern for brick construction. Wrought steel; heavy zinc coating (par. 3.10.1). Offset 2"; throw 4". Length 6"; width 7/8". Hook plate 3 1/8" x 1 1/8". Corrugated at center of "T". One pair required for each pair of blinds.

Type F1806C.—Same as type F1806, except cadmium- or zinc-plated (par. 3.9.1).

FASTENERS FOR BLINDS:

GRAVITY TYPES:



Type F1811.—For frame construction. Wrought steel; heavy zinc coating (par. 3.10.1). Automatic-action catch to hold blind open and closed. When blind is closed, it shall not be possible to unlock fastener from outside.

Type F1811C.—Wrought steel, cadmium- or zinc-plated (par. 3.9.1). Otherwise the same as type F1811.

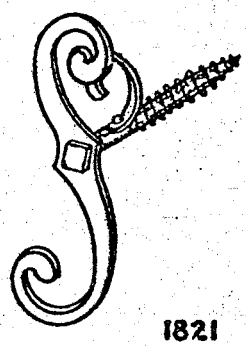
FASTENERS FOR BLINDS, CONT'D.:

GRAVITY TYPES, CONT'D.:

Type F1812.—Same as type F1811, except with back catch for driving into brick.

Type F1812C.—Wrought steel, cadmium- or zinc-plated (par. 3.9.1). Otherwise the same as type F1812.

HOLDBACKS FOR BLINDS:

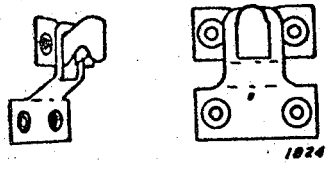


Type F1821.—Wrought steel. "S" pattern with antirattle spring and lag screw. Prime coat finish for painting. Minimum length 6".

Type F1821A.—Same as type F1821, except without spring.

SCREEN HANGERS:

FOR FULL LENGTH SCREENS:

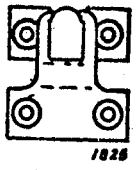


Type F1824.—Wrought steel, cadmium- or zinc-plated (par. 3.9.1). For 7/8" or 1 1/8" screens set in a 1 1/3" reveal. Eye plate 2" x 1 7/8". Hook plate 1 7/8" x 3/4". Minimum thickness of metal 0.058".

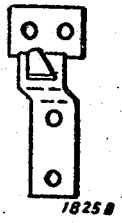
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SCREEN HANGERS, CONT'D.:

FOR FULL LENGTH SCREENS, CONT'D.:

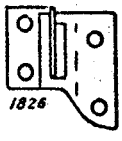


Type F1825.—For flush construction. Wrought steel, cadmium- or zinc-plated (par. 3.9.1). Eye plate 2" x 1⅞". Hook plate 1⅞" x ¾". Minimum thickness of metal 0.058".



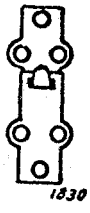
Type F1825B.—For flush construction. Wrought steel, cadmium- or zinc-plated (par. 3.9.1). Also made with varying offsets as required for screens which set in beyond face of casing. Eye plate 2$\frac{11}{16}$" x ⅞". Hook plate 1⅜" x $\frac{13}{16}$". Minimum thickness of metal 0.072".

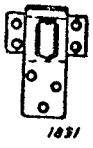
FOR HALF-LENGTH SCREENS:



Type F1826.—Wrought steel, cadmium- or zinc-plated. Eye plate 1⅝" x 1¼". Hook plate 1⅛" x $\frac{9}{16}$". Minimum thickness of metal 0.058".

STORM SASH HANGERS:



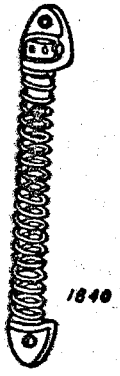


STORM SASH HANGERS, CONT'D.:

Type F1830.—For flush construction. Wrought steel, cadmium- or zinc-plated (par. 3.9.1). Eye plate 2¼" x 1¼". Hook plate 1¾" x 1¼". Minimum thickness of metal 0.120".

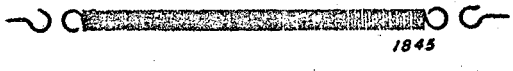
Type F1831.—For flush construction. Wrought steel, cadmium- or zinc-plated (par. 3.9.1). Eye plate 2¾" x 1". Hook plate 1⅝" x ⅞". Minimum thickness of metal 0.120".

COIL DOOR SPRINGS:



Type F1840.—Steel wire helical spring with cast-iron ends. Japanned finish. Spring tension adjustable, after application of spring, to or above the nominal strength specified below. Shall be assembled so that parts will not separate. Furnish without fastening screws unless otherwise specified.

Size, inches	9	11	13
Over-all length, inches	9	11	13
Diameter of coil, inch	¾	$\frac{15}{16}$	1
Diameter of wire, inch	0.135	0.162	0.192
Screw holes at each end	1	1	2
Nominal strength of spring at 30-inch radius, pounds	¾	1¼	2



Type F1845.—Steel wire, japanned finish. Loop ends. Two screw hooks with each spring. Length 16".

Size No.	2	3	4	5	6
Diameter of coil, inch	$\frac{5}{8}$	$\frac{11}{12}$	⅜	$\frac{17}{32}$	½
Diameter of wire, inch	0.039	0.049	0.056	0.060	0.067

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4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling and inspection.—Representative samples shall be inspected for defects and for conformity with all applicable requirements of this specification.

4.1.1 At the discretion of the Government, samples for testing will be taken at random. If the samples tested do not meet the applicable requirements, the entire lot represented thereby shall be subject to rejection.

4.1.2 In determining the minimum dimensions of bolts on cupboard, transom, and other catches or turns, it is intended that the measurements shall be made only to the nearest 1/64 inch.

4.2 Tests for electroplated coatings.

4.2.1 *Methods of test.*—Electroplated coatings shall be tested by any of the applicable methods shown below for determining thickness. The salt spray test is for determining continuity of the coatings.

4.2.2 *Umpire test.*—In case of dispute, the minimum thickness of nickel, copper, cadmium, and zinc coatings shall be determined by the microscopic test described in 4.5.

4.2.3 *Significant surfaces for tests of minimum thickness of electroplated coatings.*—In general, "significant surfaces" are those surfaces that are visible and are subjected to wear or corrosion. For this specification they are defined as those visible areas that can be touched by the surface of a sphere 0.75 inch in diameter. For delineating these areas, a brass or steel sphere 0.75 ± 0.01 inch in diameter is attached to a small rod that serves as a handle. The surface of the sphere is coated with a thin film of colored wax. When the sphere is rubbed on the surface of the fitting, areas that cannot be marked are not significant and are not subject to test.

4.3 Spot test for minimum thickness of chromium plating on nickel.

Coating	Method	Paragraph
Chromium plating on nickel	spot test	4.3
	salt spray test	4.7
Nickel plating on brass and bronze	magnetic test	4.4
	microscopic test	4.5
Nickel plating on iron and steel	magnetic test	4.4
	microscopic test	4.5
	salt spray test	4.7
Nickel plating on zinc alloy die-castings	microscopic test	4.5
	salt spray test	4.7
Cadmium plating on iron and steel	dropping test	4.6
	magnetic test	4.4
	microscopic test	4.5
Zinc coatings[1] on iron and steel	dropping test	4.6
	magnetic test	4.4
	microscopic test	4.5

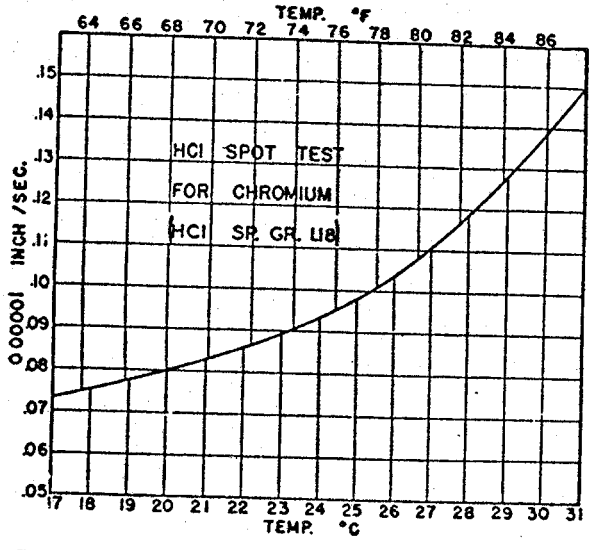
[1]The tests apply to zinc coatings applied either by electroplating or hot dipping.

4.3.1 The spot test shall be applied at one or more points on significant surfaces where the minimum thickness of chromium may be expected. At each spot to be tested, the surface shall be cleaned by light rubbing with a paste of pure magnesium oxide and water, followed by rinsing and drying. At each spot an area about 0.25 inch (6 mm.) in diameter shall be surrounded with a small ring of wax, or with a mark from a wax pencil. The article shall then be allowed to stand long enough (at least 15 minutes) to reach the room temperature, which shall be between 65° and 85°F. (18° and 30°C.) and shall be noted within 2°F. or 1°C.

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4.3.2 Reagent quality hydrochloric acid, having a specific gravity at 60°/60°F. of 1.180 ± 0.002 shall be used. (This corresponds to 11.5N ± 0.2N, which may be checked by titration.) The acid shall also be kept in the room until it acquires room temperature. One drop (0.03 to 0.05 ml.) of the hydrochloric acid shall be placed within the wax ring with a dropper. The time required from the beginning of gas evolution to the first visibility of nickel shall be determined to the nearest second with a stop watch. If the bubbling does not start at once, the surface of the chromium within the ring may be touched with a fine nickel wire to start the action.

4.3.3 The thickness of the chromium at the spot tested shall be calculated from the accompanying curve by multiplying the factor for the temperature used in the test by the number of seconds required. For example, if 16 seconds are required at 72°F.



Curve showing temperature-thickness factors for HCl spot test for chromium plating. (See 4.3.3.)

(22°C.), reference to the curve gives a factor for this temperature of 0.086 hundred thousandths of an inch per second. The thickness is therefore equal to 16 x 0.086 = 1.4 hundred-thousandths inch, or 0.000014 inch. To meet the specified minimum thickness of 0.00001 inch of chromium at any definite temperature, the following periods of time, in seconds, will be required at the temperatures indicated, with 1.180 hydrochloric acid:

Temperature		Seconds
°F	°C	
64	17.8	13.5
66	18.9	13
68	20	12.5
70	21.1	12
72	22.2	11.5
74	23.4	11
76	24.5	10.5
78	25.5	10
80	26.7	9.5
82	27.8	8.5
84	28.9	8
86	30	7.5

4.4 Magnetic test for minimum thickness of nickel, cadmium, and zinc coatings.—A properly calibrated magnetic thickness gage shall be used. The gage shall have a sensitivity of at least ± 15 percent within the range of thicknesses involved. Unless otherwise specified, the gage shall have been calibrated at the National Bureau of Standards. The thickness of the coating shall be determined at points on significant surfaces where the minimum thickness may be expected. Suitable magnetic instruments and test methods are described in National Bureau of Standards Research Papers RP994 and RP1081 by A. Brenner, published, respectively, in the Journal of Research, NBS, May 1937 (vol. 18, p. 565) and March 1938 (vol. 20, p. 357). Tests of coatings on brass and bronze are covered by RP994; tests of coatings on iron and steel by RP1081. Reprinted copies of the above papers are available from the Superintendent of Documents, Washington 25, D. C., at 10 cents per copy for RP994 and 5 cents per copy for RP1081.

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4.4.1 *Applicability of magnetic test.*—The magnetic test is regularly used only for single coatings of substantially pure metal on brass, bronze, iron, and steel. It is not normally applicable to nickel plating over copper on steel, or to nickel coatings containing 10 percent or more of cobalt when applied to brass or bronze. For such coatings, the microscopic test (par. 4.5) shall be used. Any coating of chromium over the nickel coating on brass and bronze should be entirely removed. The methods given below (par. 4.4.2) shall be used for removing chromium and for determining the presence of sufficient cobalt in the nickel coating to affect the magnetic test. In tests of nickel coatings on iron and steel, the effect of chromium and cobalt is relatively insignificant, and the above precautions ordinarily are not necessary.

4.4.2 *Preparation of nickel coatings on brass and bronze for magnetic test.*—If chromium plated, completely immerse the piece in a solution containing about 10 percent by volume of chemically pure concentrated hydrochloric acid, which has been warmed to about 50°C. (122°F.). When all chromium has been dissolved, it will be indicated by the ceasing of effervescence (usually in about 5 minutes). The piece shall then be heated in an air oven at 400°C. (790°F.) for 30 minutes and allowed to cool. This causes a yellowish tarnish of nickel coatings and a bluish tarnish of nickel deposits containing 10 percent or more of cobalt.

4.5 Microscopic test for minimum thickness of nickel, cadmium, and zinc coatings.—The microscopic test shall be made by mounting one or more cross sections cut at 90° to significant surfaces where the minimum thickness of coating may be expected. The exposed edges of the coatings shall then be polished and etched, and the thickness measured with a suitable microscope and scale.

4.6 Dropping test for minimum thickness of cadmium and zinc coatings on iron and steel.—The test shall be performed by allowing a suitable reagent to fall in drops at a definite rate on the coating at the points on significant surfaces where the minimum thickness may be expected, until the coating is penetrated. The time required for the end point to appear shall be measured with a stop watch and the room temperature observed. The thickness shall be calculated from the known characteristics of the reagent at the observed temperature, and shall be accurate within ± 15 percent. A suitable reagent and test method are described in National Bureau of Standards Research Paper RP1240 entitled "Dropping Tests for Measuring the Thickness of Zinc and Cadmium Coatings on Steel" by A. Brenner, published in the Journal of Research, NBS, September 1939 (vol. 23, p. 387). Reprinted copies of the above paper are available from the Superintendent of Documents, Washington 25, D. C., at 10 cents per copy.

4.7 Salt spray test for continuity of nickel and chromium coatings on iron, steel, and zinc alloy.—The salt spray test shall be conducted in accordance with Federal Specification QQ-M-151. After 16 hours exposure in this test, the presence on any single piece of more than 6 rust spots per square foot that are visible to the unaided eye, or of any corrosion spots larger than $\frac{1}{16}$ inch in diameter, shall be considered as "appreciable corrosion" sufficient to warrant rejection as failure to meet the test.

4.8 Tests for plastics.

4.8.1 *Test for deformation in hot water.*—Pieces for test shall be conditioned for at least 48 hours at 77° ± 2°F. and 50 ± 2 percent relative humidity, and shall then be immersed for 30 minutes in water kept at 160° ± 2°F. The pieces shall be suspended in the water by a cord or wire. Immediately after the 30-minute immersion, the items shall be examined for warping, bending, or contraction due to unmolding.

4.8.2 *Accelerated aging test.*—Pieces for test shall be heated 48 hours at a temperature of 160°F. in a circulating-air oven. After cooling to room temperature, the items shall be examined for changes in shape, color, appearance and other physical characteristics

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readily determined by handling for inspection.

4.8.3 *Test for flammability.* — Pieces for test shall be conditioned at least 48 hours at 77° ± 2°F. and 50 ± 2 percent relative humidity. Each piece taken for test shall be rigidly supported at one end with its greatest dimension horizontal. A piece of 20-mesh bunsen-burner gauze shall be supported in a horizontal position about ¼ inch under the test piece with about ½ inch of the test piece extending beyond the edge of the gauze. A bunsen burner or alcohol lamp shall be adjusted to give a flame ½ to ¾ inch in height, and the tip of the flame shall be placed so that it is just in contact with the end of the piece under test farthest from the support and at the same level. At the end of 30 seconds the flame shall be removed. If the piece does not continue to burn, the burner flame shall be again applied, immediately following extinction of the flame on the piece under test, at the same point as before. At the end of 30 seconds the burner shall be removed, and if the test piece does not continue to burn after the second ignition, it shall be reported as self-extinguishing. If burning continues, the rate of burning shall be determined as follows: Make a mark at a convenient point near the flame and a second mark a measured distance in toward the supported end of the piece. This distance should be 2 or 3 inches where practicable, but in no case less than ½ inch. Using a stop watch, determine the time required for the flame to travel from the first mark to the second, and calculate the rate of burning in inches per minute. If the shape of the article does not permit a satisfactory test, suitable pieces for test may be cut from it. Such pieces shall be rectangular in cross section and not less than 0.05 inch thick by 0.25 inch wide.

4.8.4 *Test for colorfastness to light.*— Three specimens shall be tested in accordance with method 6031 given in Federal Specification L–P–406.

4.8.5 *Test for impact strength.*—Method 1071, Izod impact strength, as given in Federal Specification L–P–406 shall be used. Ten specimens for test shall be taken from the plate in both lengthwise and crosswise directions, and shall be tested singly, in edgewise position.

4.8.6 *Boiling test.*—Three specimens 1" x 3" shall be cut from the plate and dried by heating at 50°C. for 24 hours. The weight of each specimen and its thickness at the central point shall be accurately determined. After boiling in water 2 hours, the weight and thickness shall be again determined immediately upon removal from the boiling water and quickly drying the surfaces. The change in weight and thickness shall be calculated to the nearest percent of the original dry weight and thickness. The samples shall then be examined for delamination, cracking, or splitting, and for blistering, crazing, or checking of the surfaces.

4.8.7 *Test for abrasion resistance.*—Three specimens shall be tested in accordance with method 1091, abrasion wear (loss in weight), given in Federal Specification L–P–406. For each specimen, the loss in weight shall be determined for each 1,000 revolutions for a total of 5,000 revolutions.

4.8.8 *Test for resistance to chemicals.*— Each chemical specified below shall be applied to the face of the plates in one or more spots approximately 1" in diameter, and the spots shall be covered with a petri dish. After 24 hours at room temperature (approximately 77°F.), the spot shall be washed, dried, and polished lightly with a soft, dry cloth. Careful examination shall be made for spotting and blistering. The chemicals shall be (a) trisodium phosphate, 5 percent solution; (b) phenol, 5 percent solution; and (c) carbon tetrachloride.

4.9 Tests for wood-bushed sash pulleys.

4.9.1 *Field test.*—Sample pulleys shall be selected by the inspector and the wood bushings exposed. Upon heating bushing with small flame, the impregnating material shall melt, exude from wood and collect on surface. Upon removal of flame, exuded material shall

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solidify before cooling to room temperature and at room temperature shall not be sticky, gummy, or capable of being absorbed by ordinary white blotting paper.

4.9.2 *Laboratory test.*—At the option of the inspector, wood bushings may be subjected to the following test. Bushings selected by the inspector shall be removed from the pulleys and accurately weighed. They shall then be immersed for not less than 12 hours in clean water at room temperature. Upon removal from water and immediate reweighing after quickly drying surface with absorbent material, the weight shall be not more than 5 percent greater than the original weight. The bushings shall then be placed in a desiccator containing concentrated sulphuric acid and dried for not less than 12 hours at room temperature, after which the weight shall be not more than 5 percent under the original weight. Bushings shall also withstand being maintained at a temperature of 120°F. to 125°F. with no indication of the melting or exuding of the impregnating material, and with no indication of it becoming gummy or sticky. Upon heating above its melting point, the impregnating material shall exude freely from wood and not be gummy or sticky.

5. PREPARATION FOR DELIVERY

5.1. Packaging.—Unless otherwise specified, commercial packages are acceptable under this specification.

5.2 Packing.—Unless otherwise specified, the subject commodity shall be prepared for shipment to permit acceptance by carrier for transportation at the lowest applicable rate, and to afford maximum protection from normal hazards of transportation.

5.3 Marking.—Unless otherwise specified, shipping containers shall be marked with the name of the material, the type, the size, the finish, and the quantity contained therein, as defined by the contract or order under which the shipment is made, the name of the contractor, and the number of the contract or order.

6. NOTES

6.1 Purchaser's options.—Builders' hardware should be specified by type number, and any selections should be specified as to size, hand, finish or other selective variations provided herein under each type and in 3.1 to 3.16.3, inclusive. All such options should be specified in the invitation for bids. (See 6.2.)

6.2 Types and finishes for "special application".—Types designated herein by the letters "SA" prefixed to the type number are intended for certain special applications, and may not be readily available from stock or early production. Ordinarily, these types should not be specified; when they are needed, the probable delivery date should be obtained from the source of supply before ordering. *All* types of hardware with black finishes US18, US18A, and US19 are likewise for special applications and these finishes should not be specified without first determining the probable delivery date of material so finished. Iron and steel hardware in finishes other than those listed herein in the respective types may also be subject to serious delay in time of delivery.

6.3 Finishes.

6.3.1 *Preferred finishes.*—As far as practicable, finishes should be confined to bronze (solid or plated), nickel-plated, chromium-plated, cadmium-plated, zinc-coated, and japanned. Zinc coating, of the thickness specified herein (3.10.1) is the preferred finish for steel intended for outdoor use.

6.3.2 *Selection of finishes.*—Where finishes other than those recommended above are required, selection should be confined to finishes shown in table II, which conform to standards adopted by the manufacturers as listed in the Commercial Standard for Builders' Hardware (Nontemplate), CS22–40.

6.3.3 *Standard finishes.*—The standard finishes for builders' hardware are intended primarily to promote uniformity in the appearance of builders' hardware items, and for 18 standard finishes, which are subject

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to undesirable variations in appearance, duplicate samples are used by the manufacturers as a guide for color and texture. Samples of these finishes are filed at the National Bureau of Standards for reference, and approved duplicate samples may be obtained (in full sets only) from the National Bureau of Standards at cost. (Five of these finishes have been omitted from table II because of decreasing demand.)

6.3.4 *Comparison of standard finishes.*—Owing to differences in the processes of producing hardware finishes and the variety of metals to which they are applied, it is commercially impracticable to attain an exact match. Therefore, it is understood that hardware delivered in a standard finish from two or more sources will compare reasonably when the items are viewed at arms length and approximately 2 feet apart. This approximates the conditions of use, as for example, hinges and lock trim on opposite edges of a door.

6.3.5 *Oxidized finishes* represent the natural aging of metals, and change with time; therefore, some variations from standard may be expected. An oxidized finish artificially produced may not match a similar finish produced by age.

6.4 **Dimensions.**—Where more than one dimension is given for items of builders' hardware, the vertical dimension is given first.

6.5 **Definition of "builders' hardware".**—The term "builders' hardware" is intended to cover mechanical devices for the convenience, protection, and safety of the occupants of buildings; and for supporting, guarding, operating, controlling, or securing the various movable parts of buildings, such as doors, windows, transoms, drawers, gates, and scuttles.

6.6 **New type numbers.**—Because of desirable changes in arrangement, certain items in this specification bear different type numbers from substantially equivalent items in the previous issue dated September 5, 1933. For convenience in locating such items, the corresponding type numbers are given on pages 46 and 47. However, for most of these items, as well as for items which have the same numbers as in the previous issue, there may be differences which should be carefully observed, particularly as to size and finish.

6.6.1 *Numbers in parentheses.*—New items are also provided herein which may be more suitable for some applications than items in the previous specification, and the numbers of such items are shown in parentheses.

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TABLE II. *Standard finishes for builders' hardware*

Symbol	General description	Metal applied to	Restrictions
USP	Primed for painting	Iron and steel	—
US1B	Bright japanned (black)	. . . do . . .	—
US1D	Dead black	Iron, steel, brass and bronze	—
US2C	Cadmium plated	Iron and steel	—
US2H	Zinc coated	. . . do . . .	—
US3	Bright brass[1]	Iron, steel, wrought brass, cast brass	Limited to bright or polished surfaces.
US4	Dull brass[1]	. . . do . . .	—
US5	Dull brass, oxidized and relieved[1]	. . . do . . .	Limited to ornamental designs. Plain hardware to match to be finish US4.
US9	Bright bronze[1]	Iron, steel, wrought bronze, cast bronze	Limited to bright or polished surfaces.
US10	Dull bronze[1]	. . . do . . .	—
US11	Dull bronze, oxidized and relieved[1]	. . . do . . .	Limited to ornamental designs. Plain hardware to match to be finish US10.
US14	Nickel plated[1]	Iron, steel, brass and bronze	Limited to bright or polished surfaces.
US15	Nickel plated, dull[1]	. . . do . . .	Limited to plain surfaces.
(US18)[2]	Bower-Barff[1]	Iron and steel	—
(US18A)[2]	Sanded, rust resisting black[1]	. . . do . . .	—
(US19)[2 3]	Sanded, dull black[1]	Iron, steel, brass and bronze	—
US25	White bronze[1]	Cast and wrought white bronze	Limited to white bronze compositions and to wrought white bronze in push plates, kick plates, and butts; polished surfaces.
US26	Chromium plated	Nickel coatings	Limited to bright or polished surfaces.
US26D	Chromium plated, dull[1]	. . . do . . .	Limited to plain surfaces.

[1]Standard samples are available. See 6.3.3.
[2]These black finishes are "special application". See 6.2.
[3]Finish US19 on iron and steel is the same as US18A.

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List of comparative types, FF-H-111 and FF-H-111a (see 6.6)

Old No.		*New No. in FF-H-111a*
1000	—	1125 (1125A)
1002	—	1126
1002A	—	F1126
1008	—	[SA1126][1]
1009-B1009B	—	(See Fed. Spec. FF-H-106a)
A1010	—	F1171
A1010A	—	1171
1019A	—	F1011C (1012C)
1019B	—	F1011 (F1012)
1019C	—	1011 (1012)
1020	—	1013
1022A	—	F1023AC (1023A)
1022B	—	F1023A
1025	—	1021A
1025A	—	F1021A
1027	—	1028A, 1028B
1027C	—	[SA1028C][1]
A1028	—	F1029
1044	—	1046 (1048)
1044B	—	1045 (1045R, 1048)
1047	—	1021B
1047A	—	F1021B
1050B	—	F1023B
1050A	—	F1023BC (1023B)
A1051-C1051A	—	(See Fed. Spec. FF-H-136)
1051A	—	F1051
1053A	—	1053
1060	—	1062 (1061)
1060A	—	F1062 (F1061)
1064	—	1064 (1065 A&T)
1064A	—	F1064 (F1065 A,B&T)
A1068	—	F1066A (F1066)
B1068	—	F1067 (F1067G)
B1068A	—	1067 (1067G)
1068	—	F1068
1079	—	F1075 (F1075A)
1080	—	1075
A1081	—	1070 (1071-F1078A)
B1081	—	1081
1082	—	1082
1083	—	1083
1083B	—	1085
1083C	—	F1085
1086A	—	1089 (1087-1090)
1097	—	1097 (1098)
1097A	—	F1097
1100	—	1096
1100A	—	F1096
1114	—	1114
1114A	—	F1114
1115	—	1112
1115B	—	F1112
1122	—	1122 (1123, F1123)
1134, 1134A, 1134B	—	1134 (1135, F1135, 1136, F1136)
B1137	—	F1137C
B1137A	—	F1137
1139	—	1139 (1140, 1142)
1139A	—	F1140 (F1142)
1146	—	1146
A1148	—	F1148 (F1147)
A1149	—	F1162
A1149A	—	1162 (1160-1166A)
B1149	—	1149A
B1149A	—	1149B
C1149A	—	1167 (1168, 1169)
1151	—	1156
1151A	—	F1156
1153	—	1154A (1158)
1153A	—	F1154A
1154	—	1154 (1158)
B1154A	—	F1170
1162	—	1172
1162B	—	F1172
A1162	—	F1173
A1176	—	1176

[1]See 6.2 on availability.

NOTE.—Numbers in parentheses are explained in 6.6.1.

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List of comparative types, FF-H-111 and FF-H-111a (cont'd.) (see 6.6)

Old No.		*New No. in FF-H-111a*
A1176A	—	1176A
A1187	—	1091
1189A	—	F1189 (F1188)
A1196	—	F1190C (F1190, F1195 F1195C, F1195H)
1198C	—	F1105
1199A	—	1104
1199C	—	F1104
1199, 1199B, F, G, K	—	[SA1106A-SA1106CF][1]
1208	—	1213
1208A	—	F1213
1212	—	1212
1219	—	1203 (1204, F1204, F1204A, 1205)
A1223	—	1223
A1223A	—	F1223
1224	—	1224 (1225-1227)
A1228 B1228A	—	(See Fed. Spec. FF-H-106a)
1249	—	1230B
1249A	—	1230A
1249C	—	F1232 (1232, F1233)
1250A	—	1230
1250E	—	[SA1231, SA1231A][1]
B1250	—	1240
B1250A	—	F1240 (F1243, F1245)
1262	—	1262
1264	—	1264
A1264	—	1265B
B1264	—	1265A
1266	—	1266
1269	—	1269
1269A	—	F1269
1274	—	1274
1274A	—	F1274 (1275, F1275 F1275C)
1276	—	F1276C
1276A	—	F1276
A1283	—	1280 (1280D, F1284)
1296	—	F1310
1297	—	1310
1305	—	1305 (1306-F1308)
1332	—	1322 (1323)
1332A	—	F1322 (F1323)
1334	—	1320 (1320E)
1336	—	1321 (1321E)
1340	—	1328 (1328E, 1331, F1331)
1341	—	1329 (1329E, 1330, 1330A)
1344	—	1344
1401	—	F1401E
1401A	—	F1401 (F1401C F1403, etc.)
1420	—	F1405E
1420A	—	F1405 (F1405C)
1430	—	F1430 (F1430C, D)
1510	—	F1510
1520	—	F1520 (F1520A)
1530	—	F1530
1530A	—	F1530A
1601	—	1601
1601C	—	F1601
1650	—	F1650 (F1651-F1655)
1701	—	1229F
1701A	—	1229FS
1701B	—	1229L
1701C	—	1229LS
1801	—	F1801
1801A	—	F1804
1802	—	F1805
1802A	—	F1806
1811	—	F1811 (F1811C)
1811A	—	F1812 (F1812C)
1821	—	F1821
1825	—	F1825 (F1824)
1825B	—	F1825B
1826	—	F1826
1830	—	F1830 (F1831)

[1]See 6.2 on availability.

NOTE.—Numbers in parentheses are explained in 6.6.1.

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7. DEPARTMENTAL REQUIREMENTS

7.1 The following departmental specifications, of the issues in effect on date of invitation for bids, form a part of this specification for purchases made under this specification by the respective departments.

7.2 Army.—U. S. Army Specification 100–2, Standard Specification for Marking Shipments by Contractors.

(Copies of specifications required for Army purchases may be obtained upon application to the procuring agency. Both the title and identifying symbol or number should be stipulated when requesting copies.)

7.3 Navy.

7.3.1 *Applicable specifications and other publications.*

Navy Department Specification:

General Specifications for Inspection of Material.

Army-Navy Joint Packaging Instruction:

JPI–7—Packaging and Packing of Hardware, Builders and Pole Line.

(Copies of Federal specifications and Army-Navy Joint Packaging Instructions required for Navy purchases, and Navy Department specifications, may be obtained without cost upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Center, Norfolk 11, Va. Both the title and identifying symbol or number should be stipulated when requesting copies.)

7.3.2 *Packaging, packing, and marking.*—Packaging, packing, and marking shall be in accordance with the requirements of Army-Navy Joint Packaging Instruction JPI–7.

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